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(54) **INTERFACE MODULE, INFORMATION COMMUNICATION DEVICE, AND ACTIVATION METHOD**

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CPC ***G06F 9/44505*** (2013.01)

(57) **ABSTRACT**

An interface module includes: a non-volatile storage area that stores one or more versions of firmware and a mode file; and a processor that executes a program code included in the firmware, in which each version of the firmware includes: a plurality of protocol-based program codes; and an activation unit that activates at least one of the plurality of protocol-based program codes based on description of the mode file.

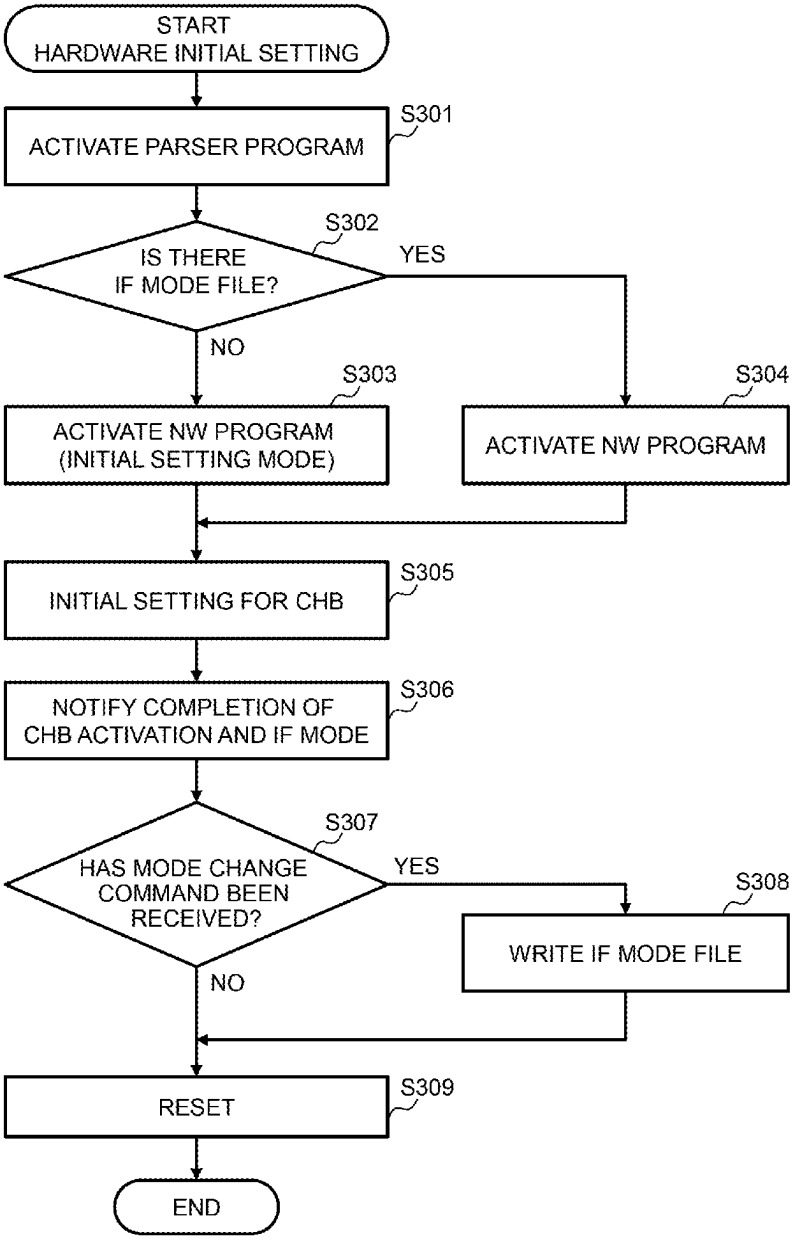

FIG. 1

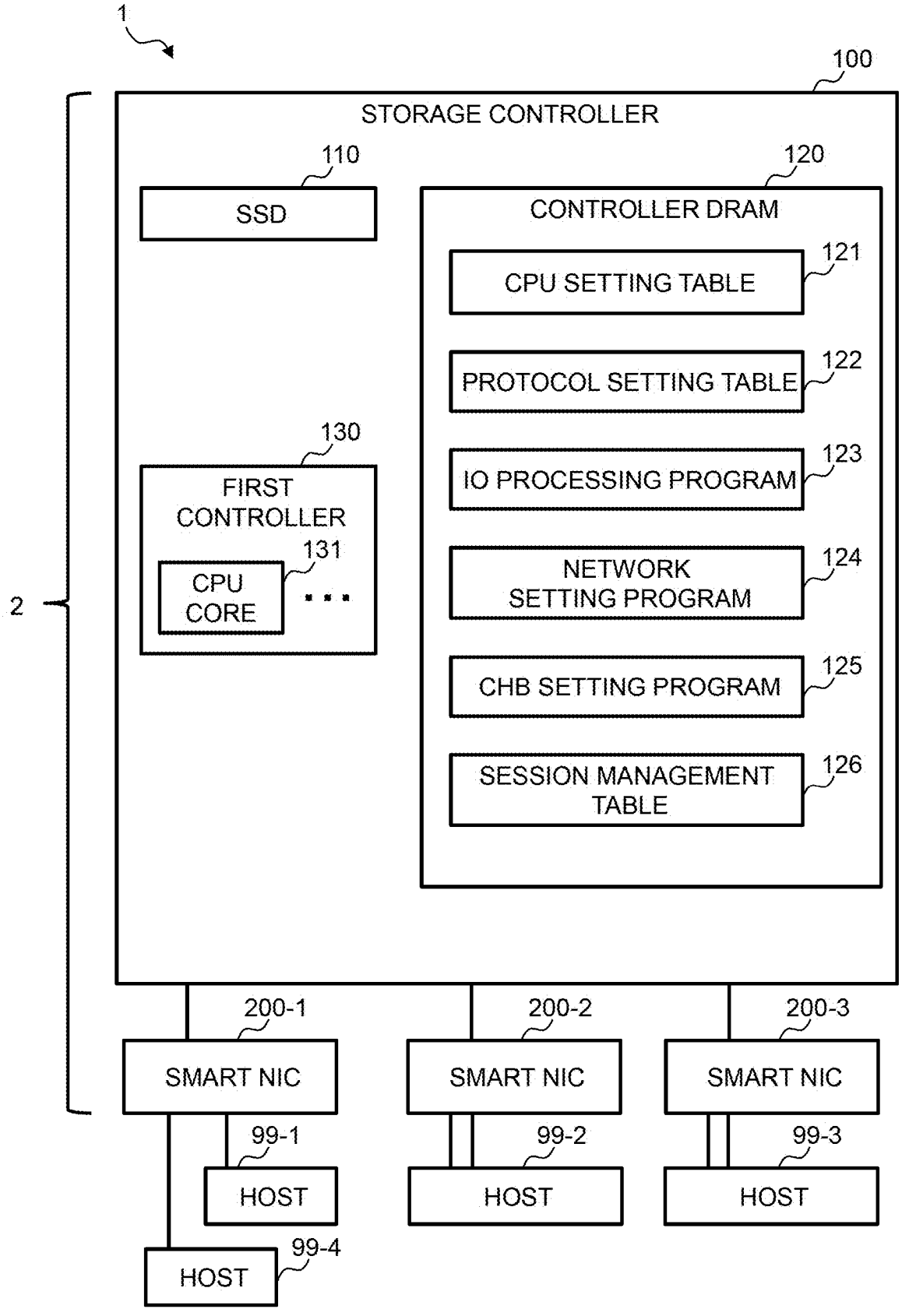
1
100
STORAGE CONTROLLER
110
SSD
120
CONTROLLER DRAM
121
CPU SETTING TABLE
122
PROTOCOL SETTING TABLE
123
IO PROCESSING PROGRAM
124
NETWORK SETTING PROGRAM
125
CHB SETTING PROGRAM
126
SESSION MANAGEMENT TABLE
130
FIRST CONTROLLER
131
CPU CORE
2
200-1
SMART NIC
200-2
SMART NIC
200-3
SMART NIC
99-1
HOST
99-2
HOST
99-3
HOST
99-4
HOST

FIG. 2

120

CONTROLLER DRAM

121

CPU SETTING TABLE

CHB ID	QUEUE HANDLING DEDICATED MODE	QUEUE HANDLING DUAL MODE	CPU RESOURCE FOR Port 0	CPU RESOURCE FOR Port 1
C1	No	Yes	50%	50%
C2	No	Yes	100%	0%
C3	Yes	No	60%	40%

122

PROTOCOL SETTING TABLE

CHB ID	iSCSI Target		iSCSI initiator		NVMe over TCP		RoCE v2	
	Port0	Port1	Port0	Port1	Port0	Port1	Port0	Port1
C1	○	-	-	-	-	○	-	-
C2	-	-	-	-	-	-	○	○
C3	-	○	-	-	○	○	-	-

126

SESSION MANAGEMENT TABLE

SESSION ID	PROTOCOL SETTING INFORMATION	NETWORK SETTING INFORMATION	LUN MAPPING INFORMATION
E1	iSCSI setting	TCP setting IP setting	#1 - 20
....	....	....	...
E100	NVMe over TCP setting	Tcp setting IP setting	#1 - 20

FIG. 3

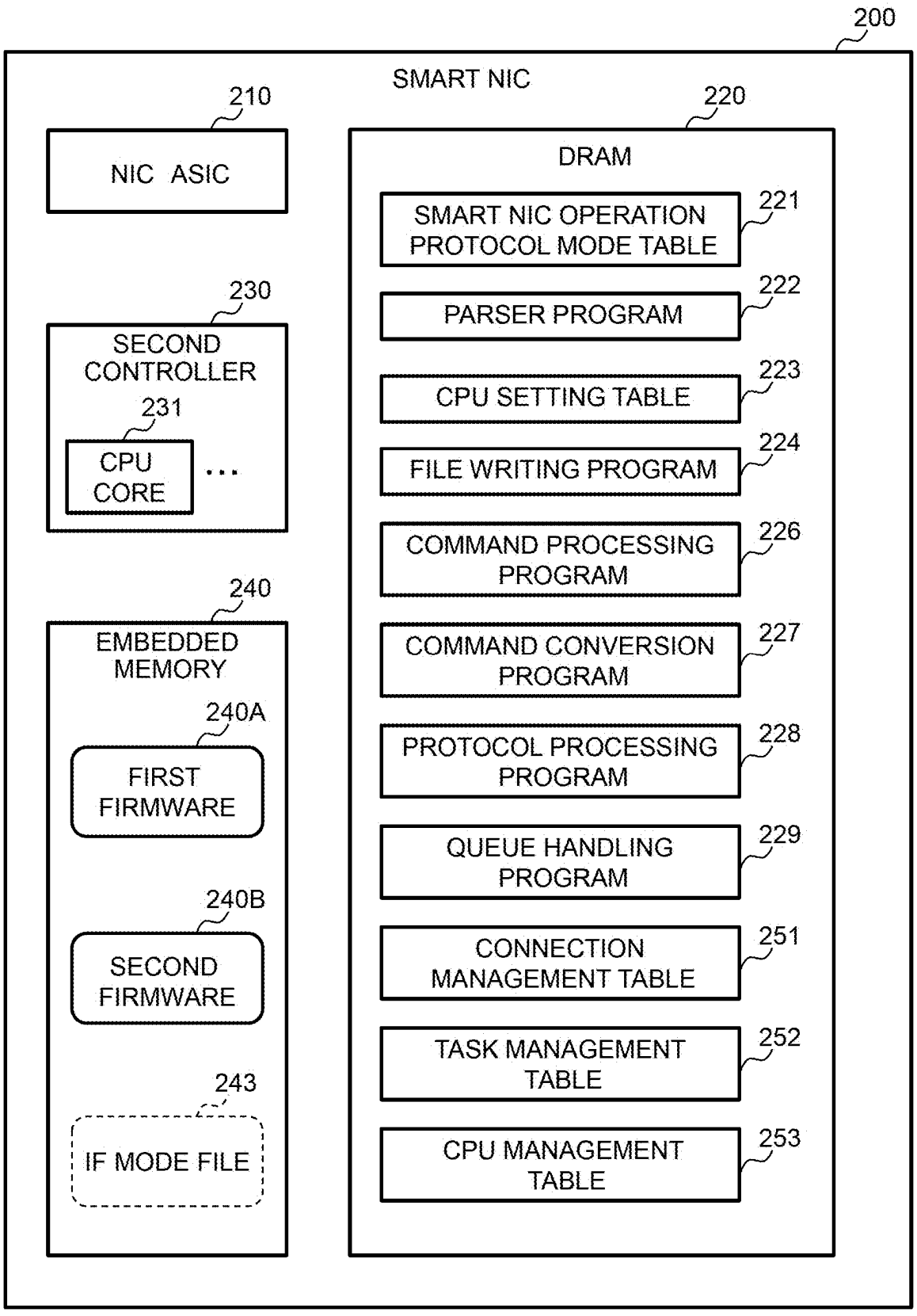
200
SMART NIC
210
NIC ASIC
220
DRAM
SMART NIC OPERATION PROTOCOL MODE TABLE
221
PARSER PROGRAM
222
CPU SETTING TABLE
223
FILE WRITING PROGRAM
224
COMMAND PROCESSING PROGRAM
226
COMMAND CONVERSION PROGRAM
227
PROTOCOL PROCESSING PROGRAM
228
QUEUE HANDLING PROGRAM
229
CONNECTION MANAGEMENT TABLE
251
TASK MANAGEMENT TABLE
252
CPU MANAGEMENT TABLE
253
230
SECOND CONTROLLER
231
CPU CORE
...
240
EMBEDDED MEMORY
240A
FIRST FIRMWARE
240B
SECOND FIRMWARE
243
IF MODE FILE

FIG. 4

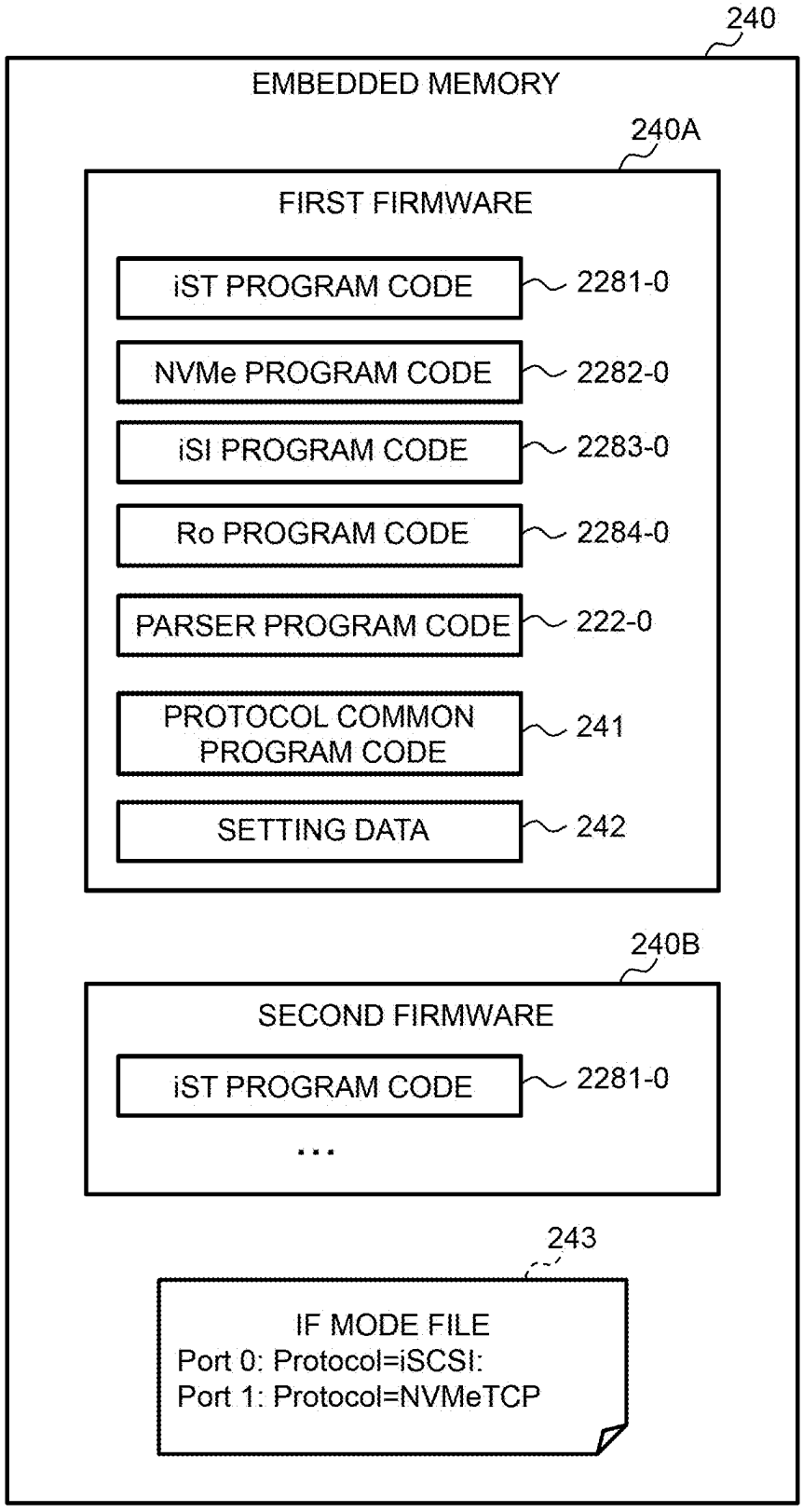
240
EMBEDDED MEMORY
240A
FIRST FIRMWARE
iST PROGRAM CODE
2281-0
NVMe PROGRAM CODE
2282-0
iSI PROGRAM CODE
2283-0
Ro PROGRAM CODE
2284-0
PARSER PROGRAM CODE
222-0
PROTOCOL COMMON PROGRAM CODE
241
SETTING DATA
242
240B
SECOND FIRMWARE
iST PROGRAM CODE
2281-0
...
243
IF MODE FILE
Port 0: Protocol=iSCSI:
Port 1: Protocol=NVMeTCP

FIG. 5

220

DRAM CHB ID: C1

221

SMART NIC OPERATION PROTOCOL MODE TABLE		
#	CONTENT OF IF MODE SELECTION FILE	IF MODE
1	Protocol=iSCSI	iSCSI Target
2	Protocol=NVMeTCP	NVMe/TCP
3	Protocol=iSCSI initiator	iSCSI initiator
4	Protocol=RoCE v2	RoCE v2
5	NO FILE EXISTS	iSCSI Target
6	OTHERS	iSCSI Target

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CPU SETTING TABLE			
Queue Handling DEDICATED MODE	Queue Handling DUAL MODE	CPU RESOURCE FOR Port 0	CPU RESOURCE FOR Port 1
No	Yes	50%	50%

FIG. 6

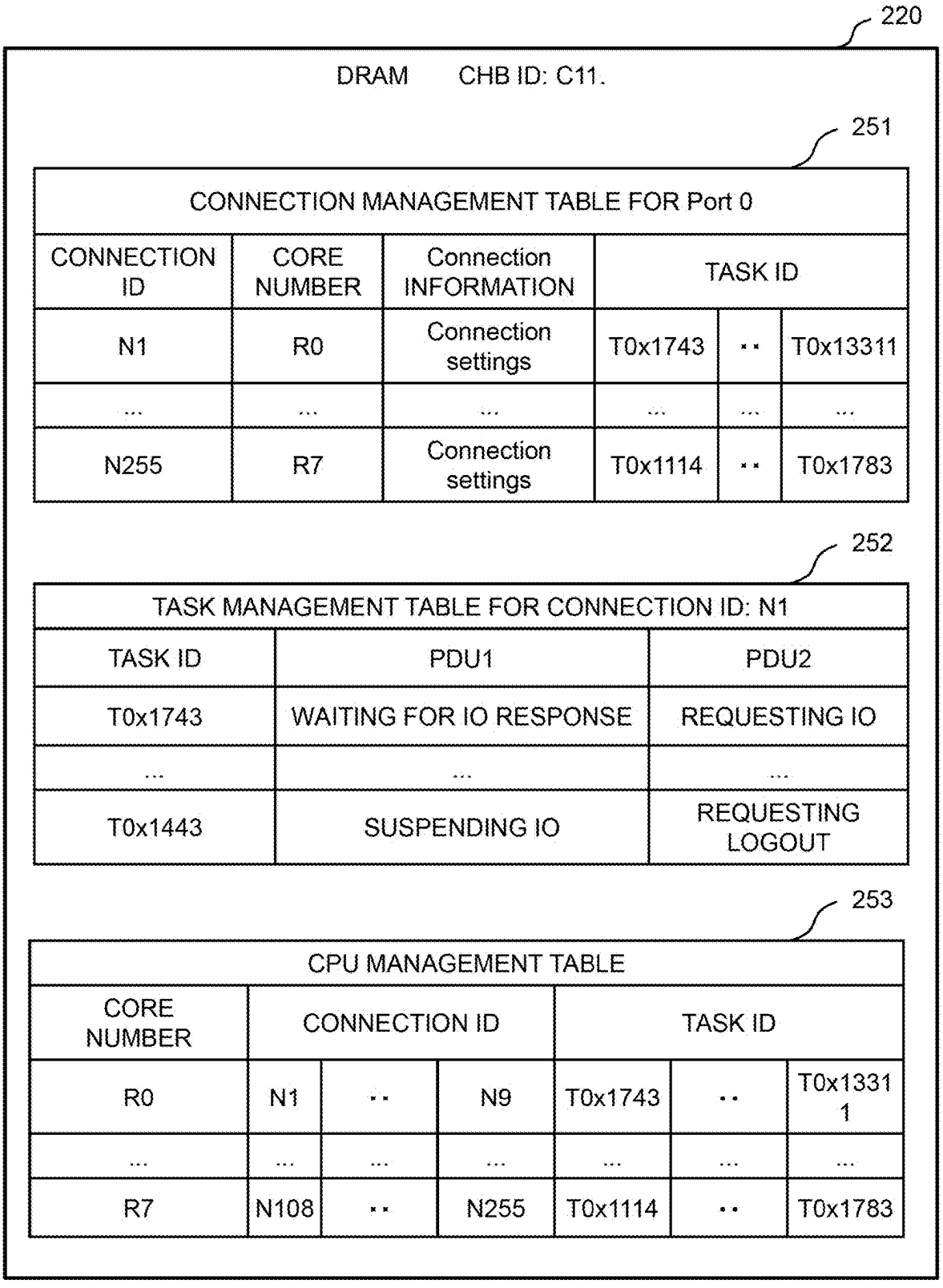

220
DRAM CHB ID: C11.
251
CONNECTION MANAGEMENT TABLE FOR Port 0
CONNECTION ID
CORE NUMBER
Connection INFORMATION
TASK ID
N1
R0
Connection settings
T0x1743
T0x13311
N255
R7
Connection settings
T0x1114
T0x1783
252
TASK MANAGEMENT TABLE FOR CONNECTION ID: N1
TASK ID
PDU1
PDU2
T0x1743
WAITING FOR IO RESPONSE
REQUESTING IO
T0x1443
SUSPENDING IO
REQUESTING LOGOUT
253
CPU MANAGEMENT TABLE
CORE NUMBER
CONNECTION ID
TASK ID
R0
N1
N9
T0x1743
T0x13311
R7
N108
N255
T0x1114
T0x1783

FIG. 7

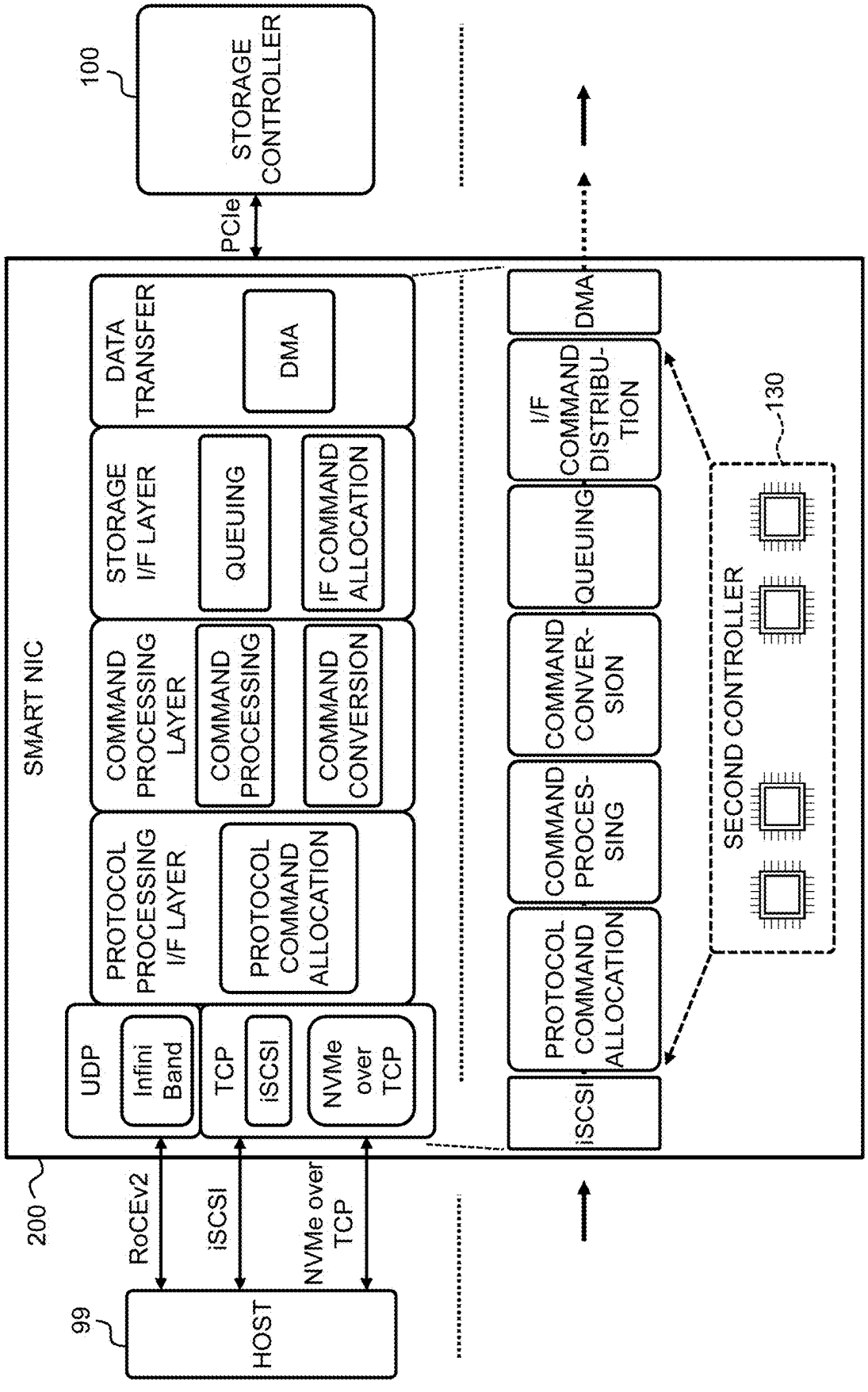
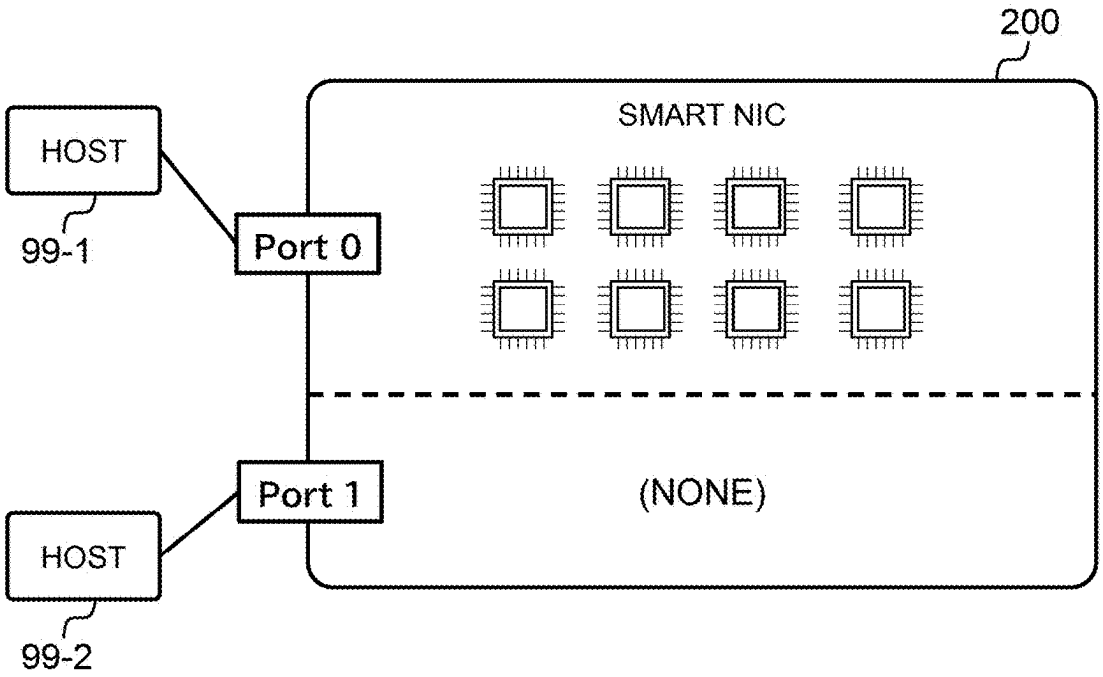

100
STORAGE CONTROLLER
PCIe
SMART NIC
DATA TRANSFER
DMA
STORAGE I/F LAYER
QUEUING
IF COMMAND ALLOCATION
COMMAND PROCESSING LAYER
COMMAND PROCESSING
COMMAND CONVERSION
PROTOCOL PROCESSING I/F LAYER
PROTOCOL COMMAND ALLOCATION
UDP
Infini Band
TCP
iSCSI
NVMe over TCP
I/F COMMAND DISTRIBU-TION
DMA
QUEUING
COMMAND CONVER-SION
COMMAND PROCES-SING
PROTOCOL COMMAND ALLOCATION
iSCSI
130
SECOND CONTROLLER
200
RoCEv2
iSCSI
NVMe over TCP
99
HOST

FIG. 8


200
SMART NIC
HOST
99-1
Port 0
Port 1
(NONE)
HOST
99-2

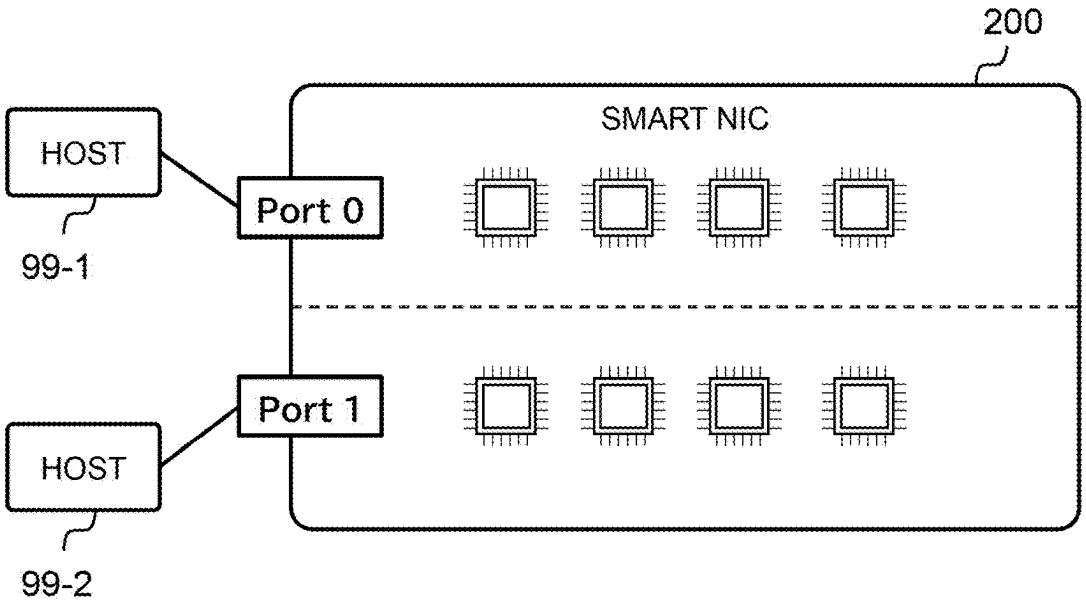

200
SMART NIC
HOST
99-1
Port 0
Port 1
HOST
99-2

FIG. 9

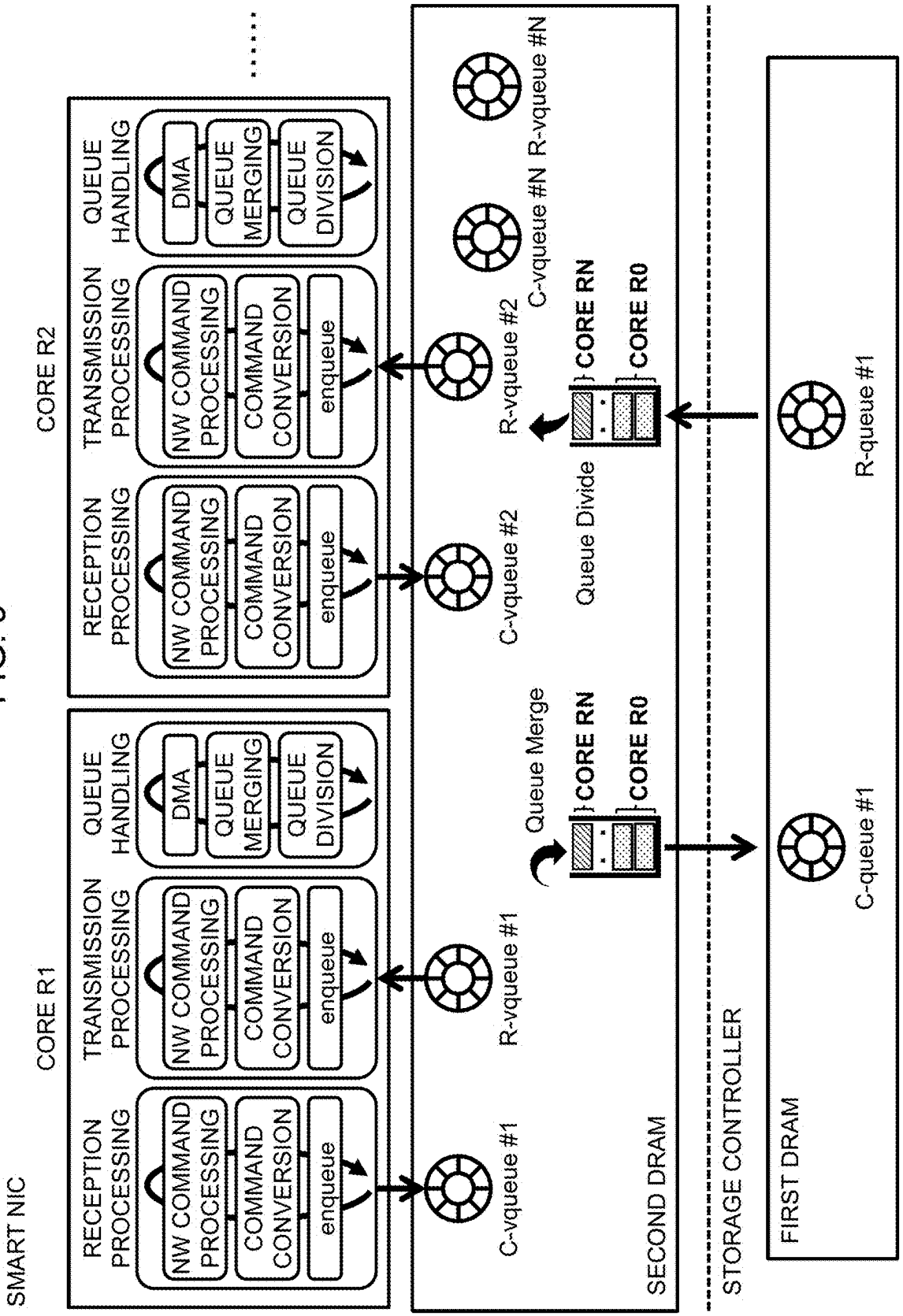

SMART NIC
CORE R1
CORE R2
RECEPTION PROCESSING
NW COMMAND PROCESSING
COMMAND CONVERSION
enqueue
TRANSMISSION PROCESSING
QUEUE HANDLING
DMA
QUEUE MERGING
QUEUE DIVISION
C-vqueue #1
C-vqueue #2
R-vqueue #1
R-vqueue #2
C-vqueue #N
R-vqueue #N
Queue Merge
Queue Divide
CORE RN
CORE R0
SECOND DRAM
STORAGE CONTROLLER
FIRST DRAM
C-queue #1
R-queue #1

FIG. 10

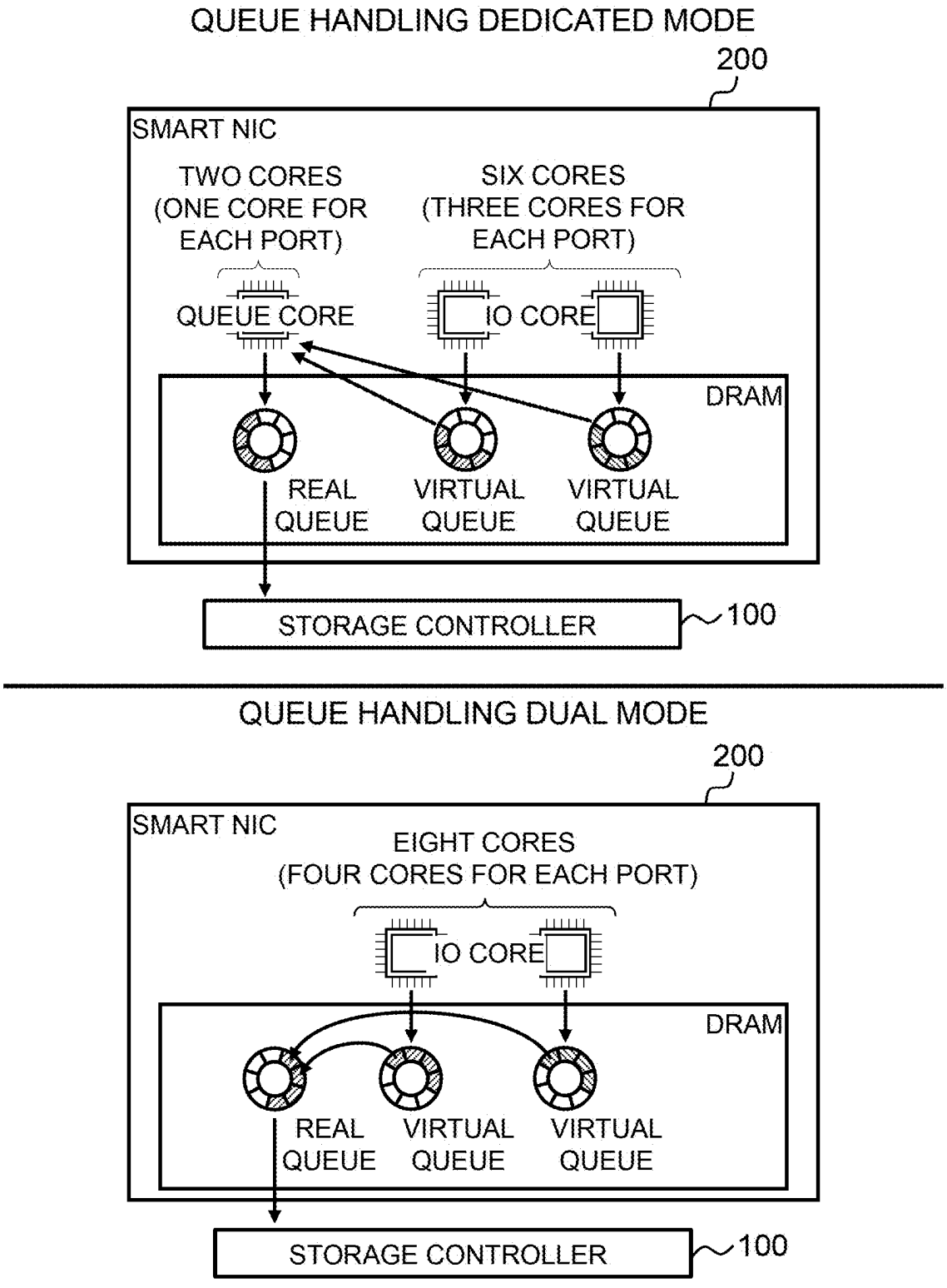

QUEUE HANDLING DEDICATED MODE
200
SMART NIC
TWO CORES
(ONE CORE FOR
EACH PORT)
SIX CORES
(THREE CORES FOR
EACH PORT)
QUEUE CORE
IO CORE
DRAM
REAL
QUEUE
VIRTUAL
QUEUE
VIRTUAL
QUEUE
STORAGE CONTROLLER
100
QUEUE HANDLING DUAL MODE
200
SMART NIC
EIGHT CORES
(FOUR CORES FOR EACH PORT)
IO CORE
DRAM
REAL
QUEUE
VIRTUAL
QUEUE
VIRTUAL
QUEUE
STORAGE CONTROLLER
100

FIG. 11

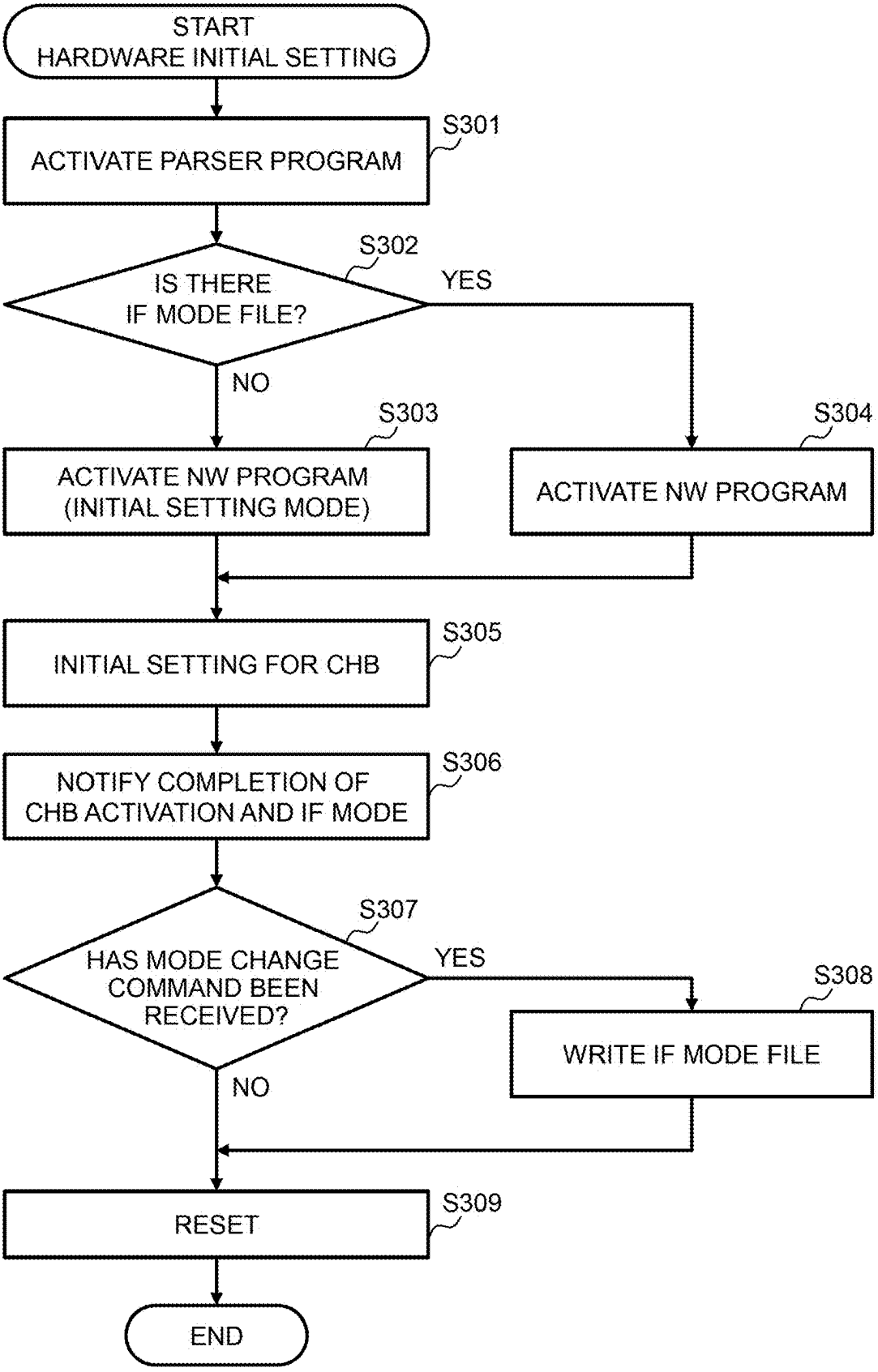

START
HARDWARE INITIAL SETTING
ACTIVATE PARSER PROGRAM
S301
IS THERE
IF MODE FILE?
S302
YES
NO
ACTIVATE NW PROGRAM
(INITIAL SETTING MODE)
S303
ACTIVATE NW PROGRAM
S304
INITIAL SETTING FOR CHB
S305
NOTIFY COMPLETION OF
CHB ACTIVATION AND IF MODE
S306
HAS MODE CHANGE
COMMAND BEEN
RECEIVED?
S307
YES
NO
WRITE IF MODE FILE
S308
RESET
S309
END

FIG. 12

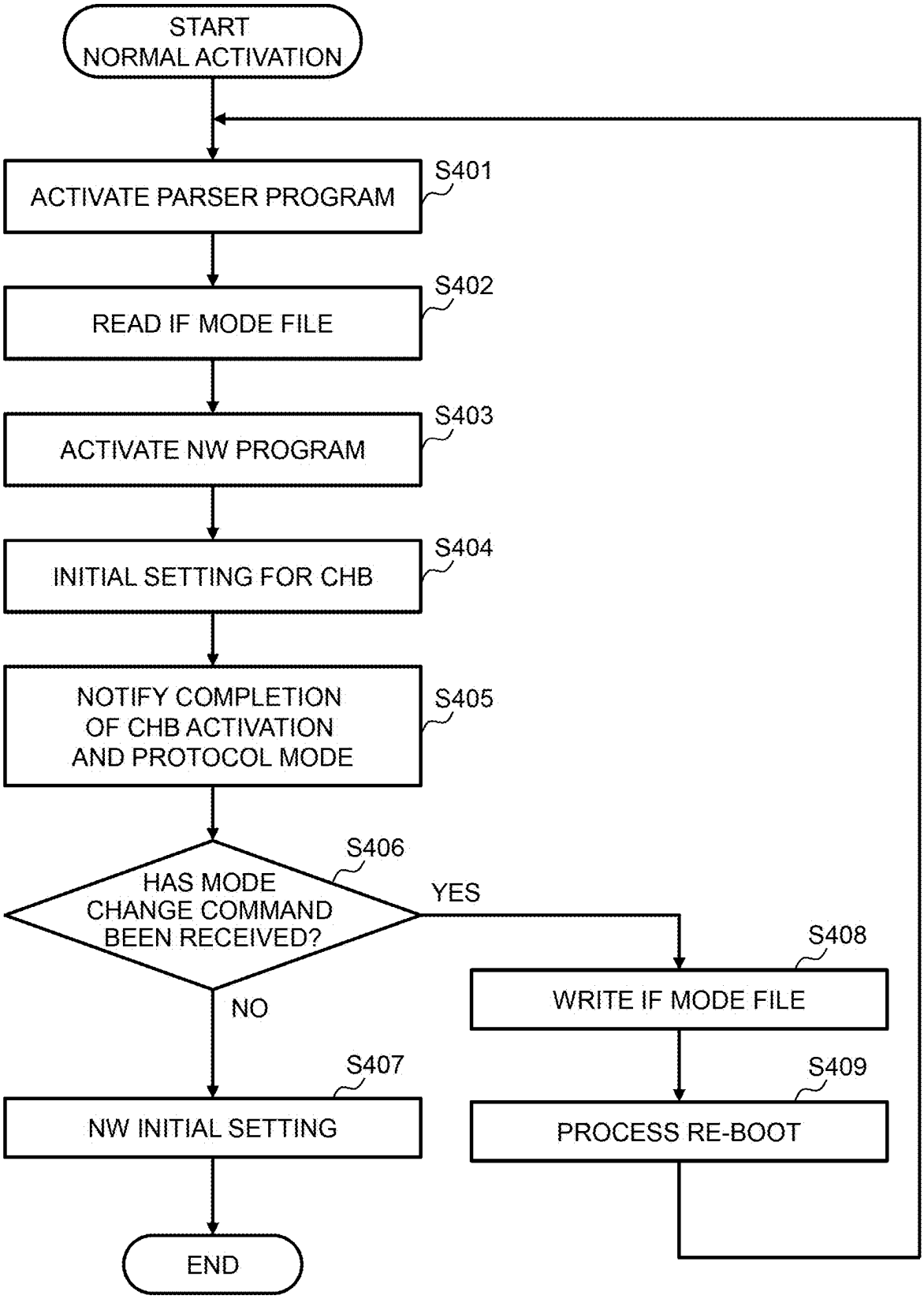

START
NORMAL ACTIVATION
ACTIVATE PARSER PROGRAM
S401
READ IF MODE FILE
S402
ACTIVATE NW PROGRAM
S403
INITIAL SETTING FOR CHB
S404
NOTIFY COMPLETION
OF CHB ACTIVATION
AND PROTOCOL MODE
S405
HAS MODE
CHANGE COMMAND
BEEN RECEIVED?
S406
YES
NO
WRITE IF MODE FILE
S408
PROCESS RE-BOOT
S409
NW INITIAL SETTING
S407
END

FIG. 13

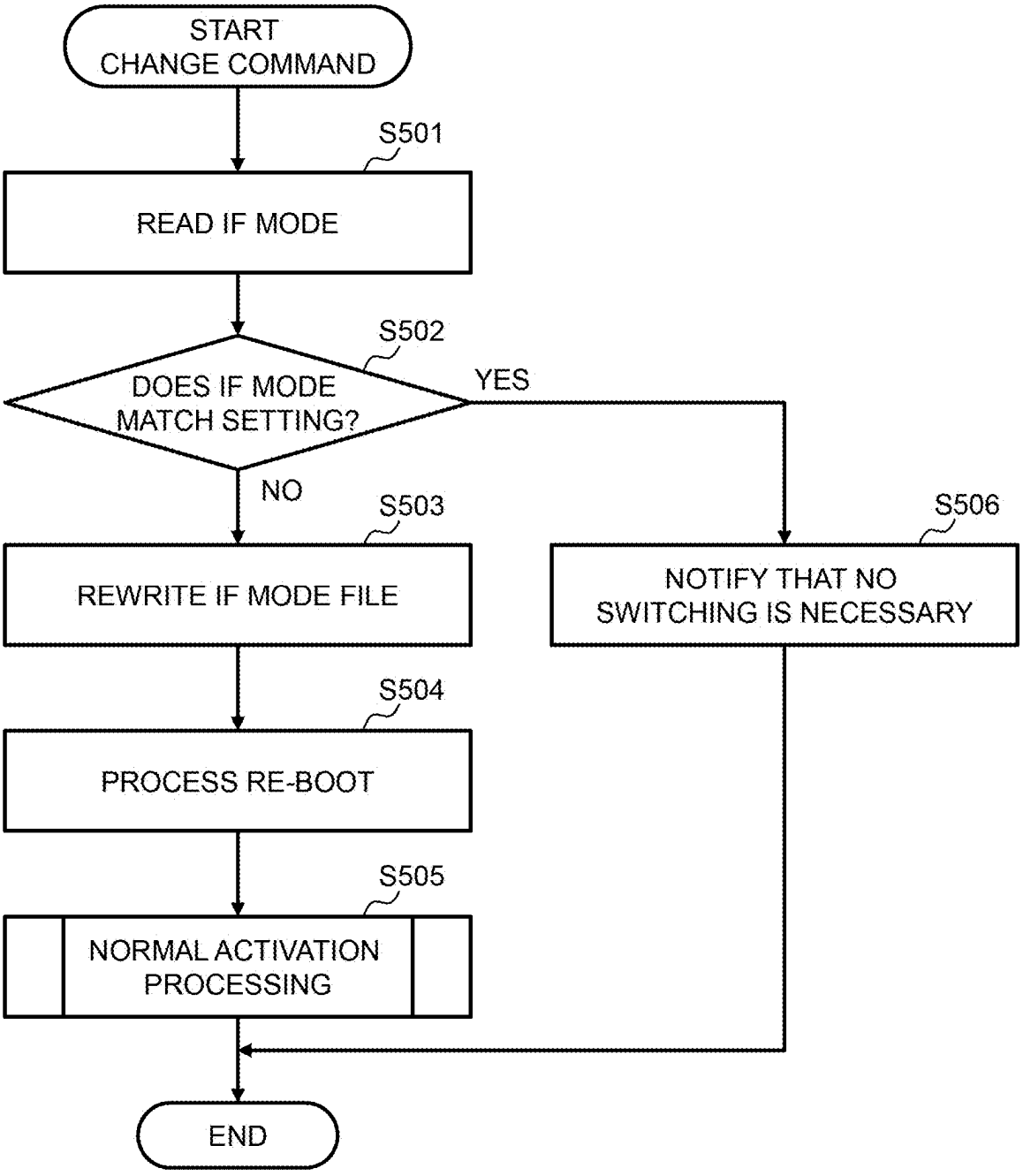

START
CHANGE COMMAND
S501
READ IF MODE
S502
DOES IF MODE
MATCH SETTING?
YES
NO
S503
REWRITE IF MODE FILE
S504
PROCESS RE-BOOT
S505
NORMAL ACTIVATION
PROCESSING
S506
NOTIFY THAT NO
SWITCHING IS NECESSARY
END

FIG. 14

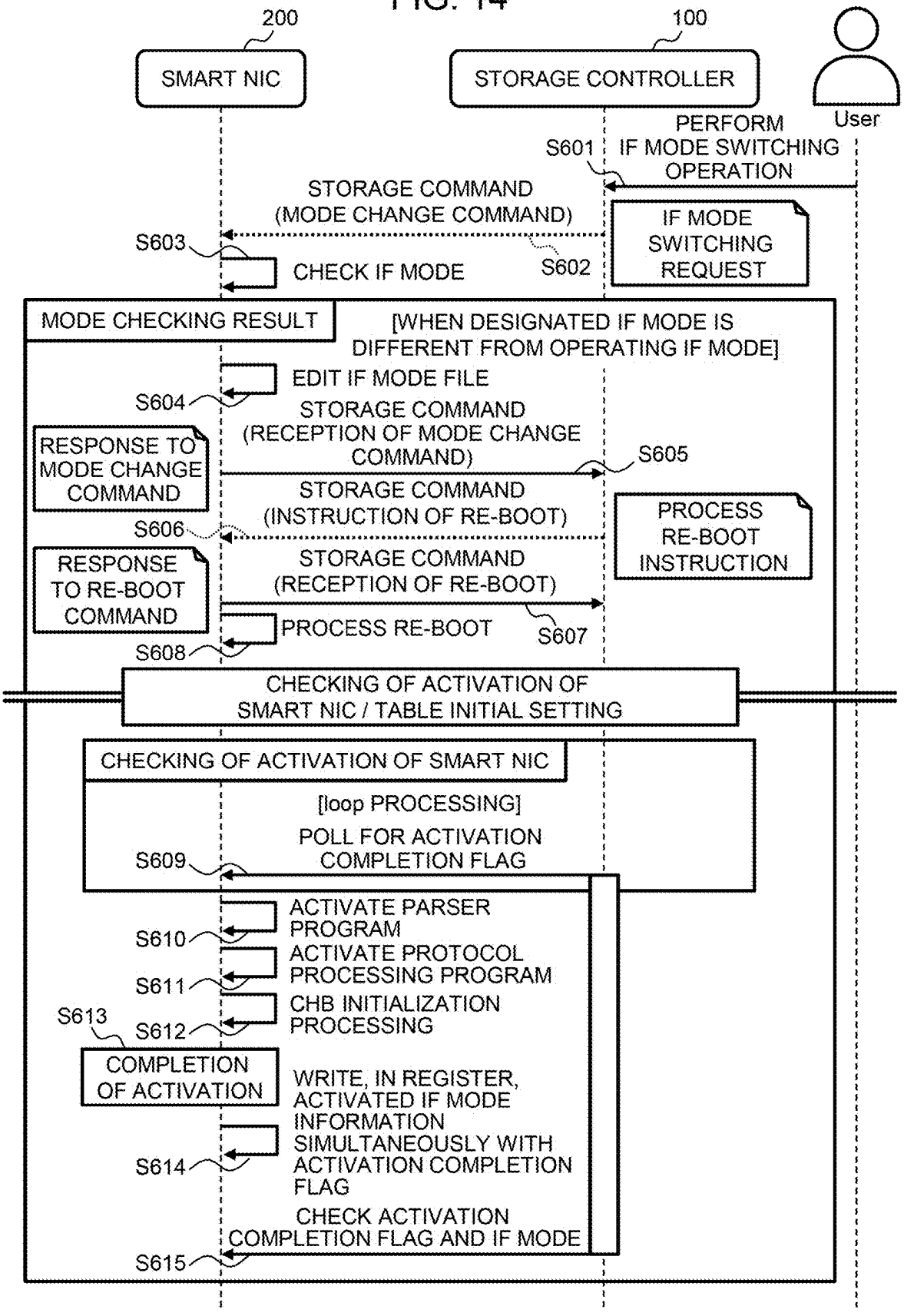

200
SMART NIC
100
STORAGE CONTROLLER
User
PERFORM IF MODE SWITCHING OPERATION
S601
STORAGE COMMAND (MODE CHANGE COMMAND)
S602
IF MODE SWITCHING REQUEST
S603
CHECK IF MODE
MODE CHECKING RESULT
[WHEN DESIGNATED IF MODE IS DIFFERENT FROM OPERATING IF MODE]
EDIT IF MODE FILE
S604
STORAGE COMMAND (RECEPTION OF MODE CHANGE COMMAND)
S605
RESPONSE TO MODE CHANGE COMMAND
STORAGE COMMAND (INSTRUCTION OF RE-BOOT)
S606
PROCESS RE-BOOT INSTRUCTION
RESPONSE TO RE-BOOT COMMAND
STORAGE COMMAND (RECEPTION OF RE-BOOT)
S607
PROCESS RE-BOOT
S608
CHECKING OF ACTIVATION OF SMART NIC / TABLE INITIAL SETTING
CHECKING OF ACTIVATION OF SMART NIC
[loop PROCESSING]
POLL FOR ACTIVATION COMPLETION FLAG
S609
ACTIVATE PARSER PROGRAM
S610
ACTIVATE PROTOCOL PROCESSING PROGRAM
S611
CHB INITIALIZATION PROCESSING
S612
S613
COMPLETION OF ACTIVATION
WRITE, IN REGISTER, ACTIVATED IF MODE INFORMATION SIMULTANEOUSLY WITH ACTIVATION COMPLETION FLAG
S614
CHECK ACTIVATION COMPLETION FLAG AND IF MODE
S615

FIG. 15

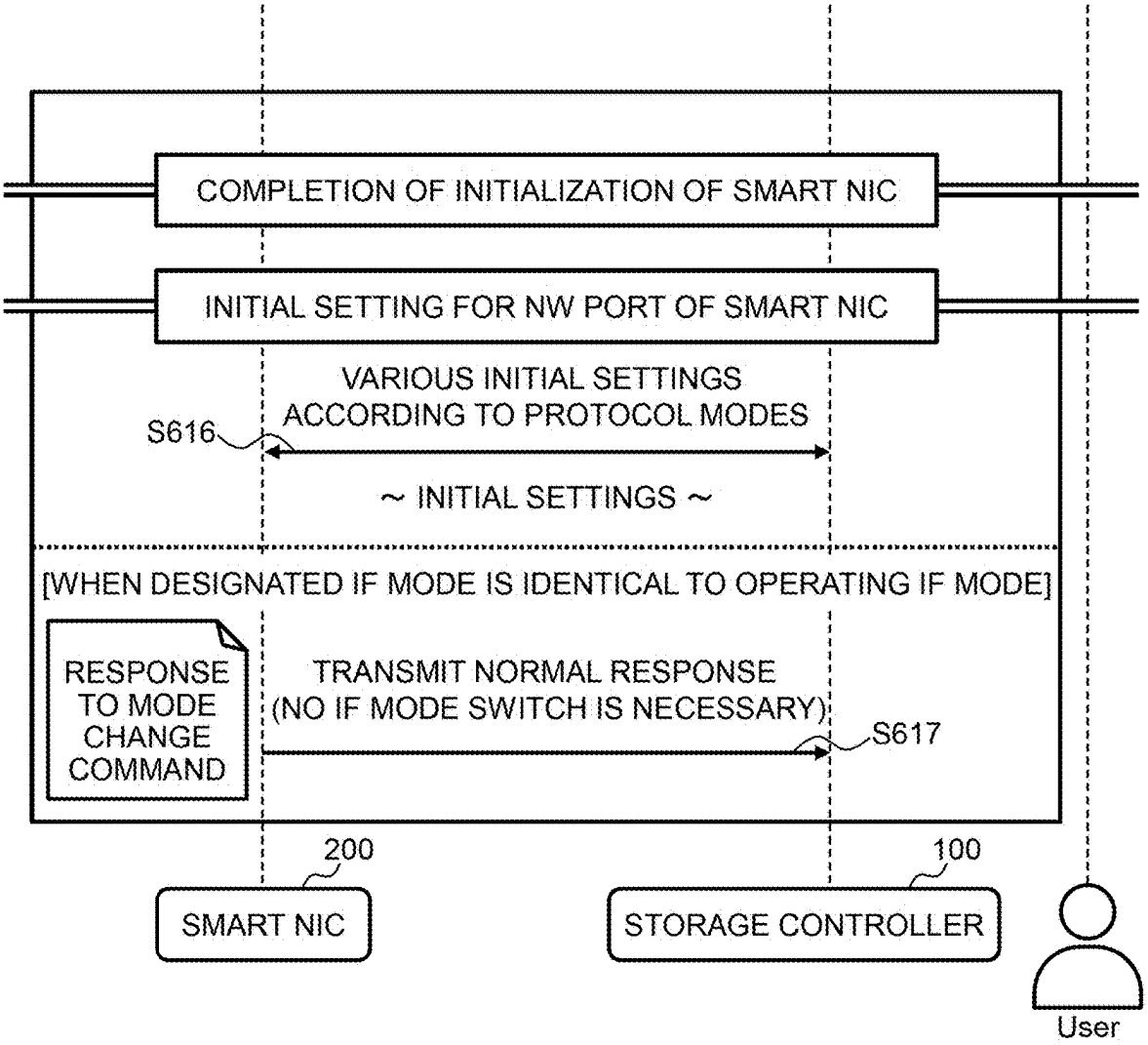

COMPLETION OF INITIALIZATION OF SMART NIC
INITIAL SETTING FOR NW PORT OF SMART NIC
VARIOUS INITIAL SETTINGS ACCORDING TO PROTOCOL MODES
S616
~ INITIAL SETTINGS ~
[WHEN DESIGNATED IF MODE IS IDENTICAL TO OPERATING IF MODE]
RESPONSE TO MODE CHANGE COMMAND
TRANSMIT NORMAL RESPONSE (NO IF MODE SWITCH IS NECESSARY)
S617
200
100
SMART NIC
STORAGE CONTROLLER
User

FIG. 16

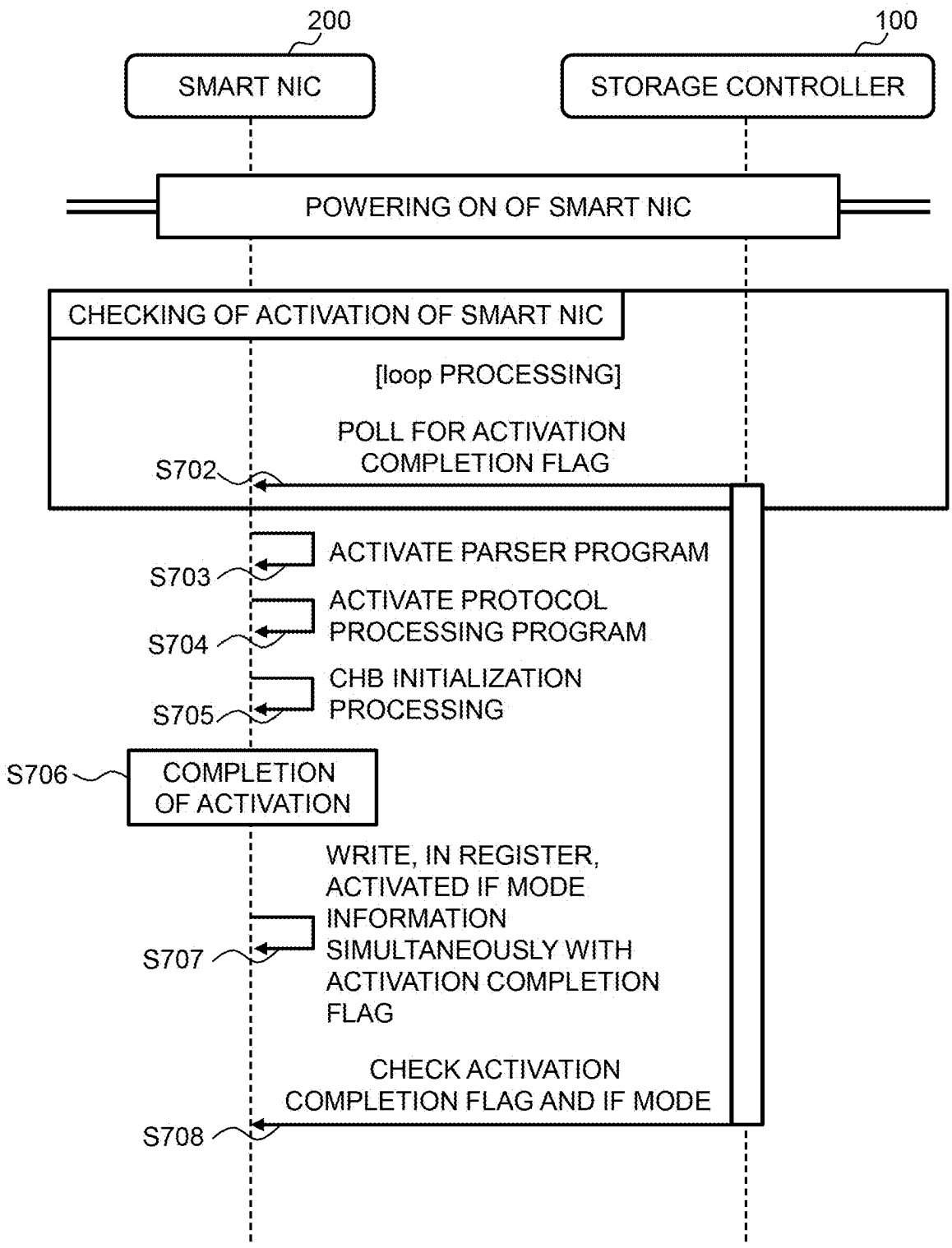

200
100
SMART NIC
STORAGE CONTROLLER
POWERING ON OF SMART NIC
CHECKING OF ACTIVATION OF SMART NIC
[loop PROCESSING]
POLL FOR ACTIVATION COMPLETION FLAG
S702
ACTIVATE PARSER PROGRAM
S703
ACTIVATE PROTOCOL PROCESSING PROGRAM
S704
CHB INITIALIZATION PROCESSING
S705
S706
COMPLETION OF ACTIVATION
WRITE, IN REGISTER, ACTIVATED IF MODE INFORMATION SIMULTANEOUSLY WITH ACTIVATION COMPLETION FLAG
S707
CHECK ACTIVATION COMPLETION FLAG AND IF MODE
S708

FIG. 17

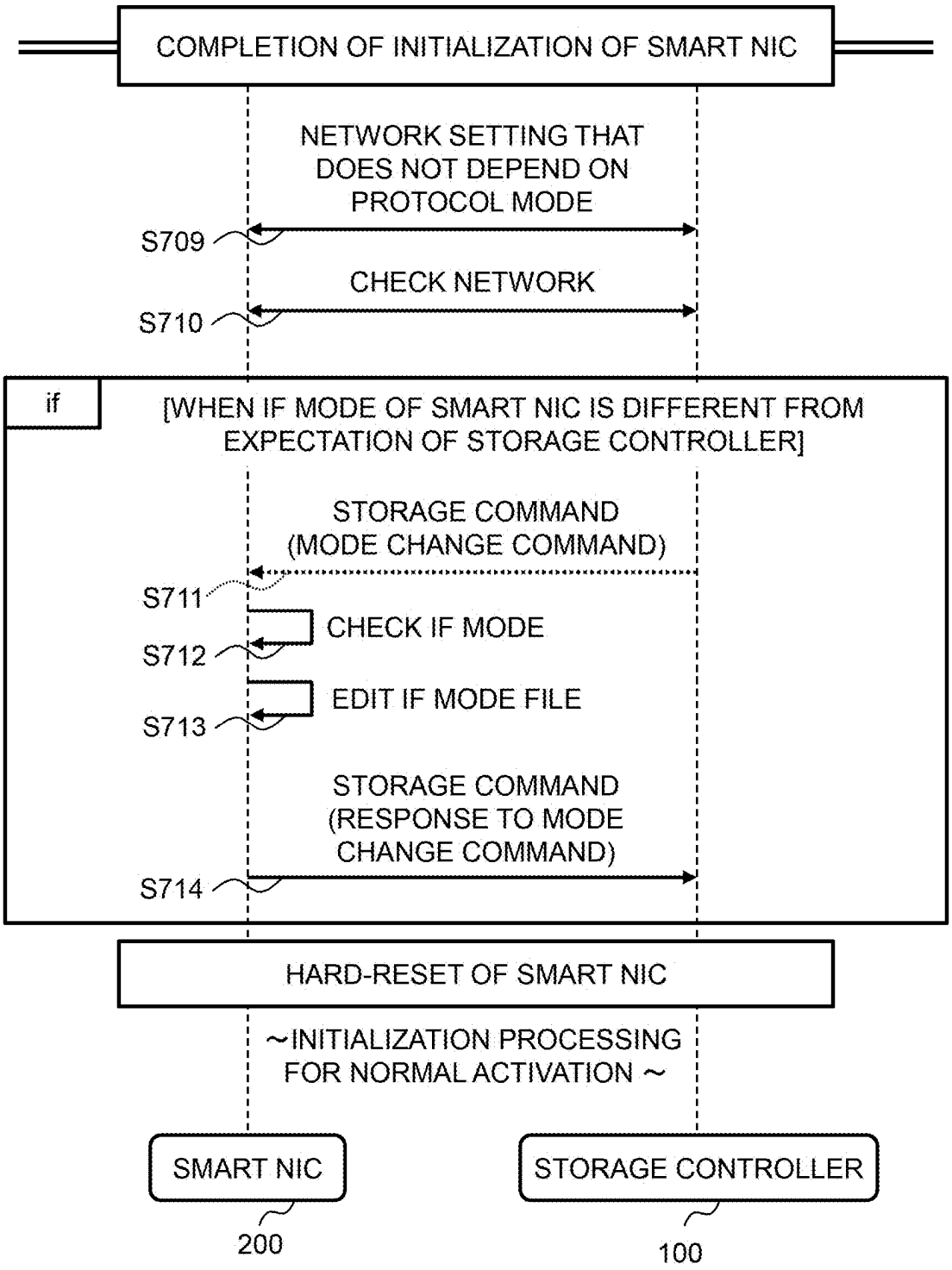
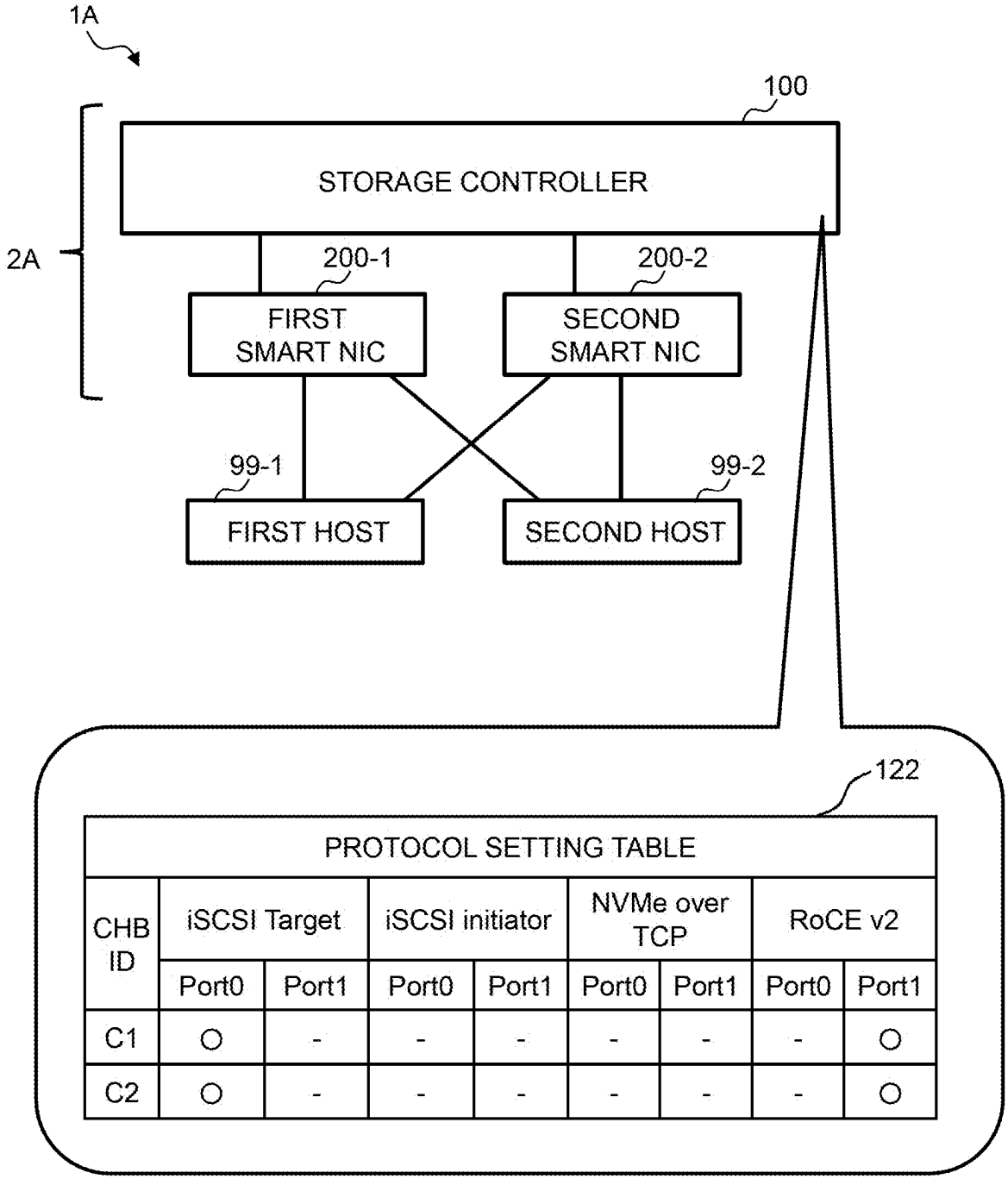

COMPLETION OF INITIALIZATION OF SMART NIC
NETWORK SETTING THAT DOES NOT DEPEND ON PROTOCOL MODE
S709
CHECK NETWORK
S710
if
[WHEN IF MODE OF SMART NIC IS DIFFERENT FROM EXPECTATION OF STORAGE CONTROLLER]
STORAGE COMMAND (MODE CHANGE COMMAND)
S711
CHECK IF MODE
S712
EDIT IF MODE FILE
S713
STORAGE COMMAND (RESPONSE TO MODE CHANGE COMMAND)
S714
HARD-RESET OF SMART NIC
~INITIALIZATION PROCESSING FOR NORMAL ACTIVATION ~
SMART NIC
200
STORAGE CONTROLLER
100

FIG. 18



PROTOCOL SETTING TABLE

CHB ID	iSCSI Target		iSCSI initiator		NVMe over TCP		RoCE v2	
	Port0	Port1	Port0	Port1	Port0	Port1	Port0	Port1
C1	○	-	-	-	-	-	-	○
C2	○	-	-	-	-	-	-	○

INTERFACE MODULE, INFORMATION COMMUNICATION DEVICE, AND ACTIVATION METHOD

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Japanese Patent Application No. 2024-034188, filed Mar. 6, 2024, the contents of which are incorporated herein by reference in its entirety for all purposes.

BACKGROUND OF THE INVENTION

1. Technical Field

[0002] The present invention relates to an interface module, an information communication device, and an activation method.

2. Description of the Related Art

[0003] Since network protocols used by storages are being newly developed day by day, the storages need to support various network protocols. Conventionally, since an available network protocol is fixed for each channel board (hereafter referred to as "CHB"), it has been necessary to replace the CHB or reconnect a network cable in order to change the network protocol used by the storage. In recent years, a CHB capable of changing a network protocol has also been developed. By using this CHB, there is no need to replace the CHB or reconnect the network cable. For example, techniques disclosed in JP 2015-32304 A and JP 2007-513436 A have been known regarding a storage supporting a plurality of protocols.

SUMMARY OF THE INVENTION

[0004] Even though any of JP 2015-32304 A and JP 2007-513436 A is applied, it is not possible to designate a network protocol to be set in advance at the time of activating the CHB. Therefore, when the desired network protocol is not set at the time of activating the CHB, it is necessary to designate an NW protocol and re-activate the CHB, and there is a problem that the time until the storage becomes available becomes long.

[0005] An interface module according to a first aspect of the present invention is an interface module including: a non-volatile storage area that stores one or more versions of firmware and a mode file; and a processor that executes a program code included in the firmware, in which each version of the firmware includes: a plurality of protocol-based program codes; and an activation unit that activates at least one of the plurality of protocol-based program codes based on description of the mode file.

[0006] An information communication device including: a first interface module that is the aforementioned interface module; and a storage controller that receives at least one of a read command and a write command via the first interface module.

[0007] An activation method according to a third aspect of the present invention is an activation method executed by an interface module including: a non-volatile storage area that stores one or more versions of firmware and a mode file; and a processor that executes a program code included in the firmware, each version of the firmware including: a plurality of protocol-based program codes; and an activation unit that activates at least one of the plurality of protocol-based program codes based on description of the mode file, the activation method including: the processor operating the activation unit of any one version of the firmware to activate at least one of the plurality of protocol-based program codes based on the description of the mode file.

[0008] According to the present invention, since it is not necessary to switch a protocol by firmware exchange while supporting a plurality of network protocols, it is possible to minimize downtime. In addition, since it is not necessary to prepare separate firmware for each protocol, not only hardware but also software can be unified.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is an overall configuration diagram of a storage system including a smart NIC;

[0010] FIG. **2** is a diagram illustrating examples of a CPU setting table, a protocol setting table, and a session management table;

[0011] FIG. **3** is a configuration diagram of the smart NIC;

[0012] FIG. **4** is a detailed configuration diagram of an embedded memory;

[0013] FIG. **5** is a diagram illustrating examples of a smart NIC operation protocol mode table and a CPU setting table;

[0014] FIG. **6** is a diagram illustrating examples of a connection management table, a task management table, and a CPU management table;

[0015] FIG. **7** is a diagram illustrating an outline of processing executed by the smart NIC;

[0016] FIG. **8** is a diagram illustrating an example of allocation of CPU cores in the smart NIC;

[0017] FIG. **9** is a schematic diagram illustrating processing for each core of the smart NIC;

[0018] FIG. **10** is a diagram for explaining a queue handling dedicated mode and a queue handling dual mode;

[0019] FIG. **11** is a flowchart illustrating processing at the time of hardware initial setting of the smart NIC;

[0020] FIG. **12** is a flowchart illustrating processing at the time of normal activation of the smart NIC;

[0021] FIG. **13** is a flowchart illustrating processing of the smart NIC when an IF mode of the smart NIC is switched by a user;

[0022] FIG. **14** is a time chart illustrating processing of the storage system when the IF mode of the smart NIC is switched by the user;

[0023] FIG. **15** is a time chart illustrating processing of the storage system when the IF mode of the smart NIC is switched by the user;

[0024] FIG. **16** is a time chart illustrating processing at the time of hardware initial setting of the smart NIC;

[0025] FIG. **17** is a time chart illustrating processing at the time of hardware initial setting of the smart NIC; and

[0026] FIG. **18** is an overall configuration diagram of a storage system according to a second embodiment.

DETAILED DESCRIPTION

FIRST EMBODIMENT

[0027] Hereinafter, a smart NIC that is an interface module according to a first embodiment will be described with reference to FIGS. **1** to **16**. In the present embodiment, a network interface card is also referred to as "NIC", an

interface is also referred to as "IF", a network is also referred to as "NW", and a channel board is also referred to as "CHB".

[0028] FIG. **1** is an overall configuration diagram of a storage system **1** including a smart NIC **200**. The storage system **1** includes a storage controller **100**, at least one smart NIC **200**, and at least one host **99**. In FIG. **1**, branch numbers are assigned to identify the plurality of smart NICs **200** and the plurality of hosts. The smart NICs **200** have a common configuration to the extent described below.

[0029] Although the storage controller **100** is connected to three smart NICs **200** in FIG. **1**, the number of smart NICs **200** connected to the storage controller **100** may be one or more. The storage controller **100** identifies each smart NIC **200** by a channel board ID. The channel board ID is an identifier of a channel board, that is, a smart NIC **200**, and is expressed by a combination of "C" and a number in the present embodiment. That is, channel board IDs "C1", "C2", and "C3" are set to the three illustrated smart NICs **200**.

[0030] The smart NIC **200** is connected to the storage controller **100** according to a prescribed communication standard, for example, PICe. Communication is performed between the smart NIC **200** and the host **99** according to a communication protocol that is publicly known or that will be developed in the future. The communication protocol is iSCSI, Infini Band, NVMe over TCP, RoCE V2, or the like. The communication protocol used for communication between the smart NIC **200** and the host **99** can be changed. This will be described in detail below.

[0031] The storage controller **100** includes an SSD **110**, a controller DRAM **120**, and a first controller **130**. The SSD **110** is a non-volatile memory using a semiconductor, that is, a solid state drive. Note that the SSD21 is merely described as an example of a non-volatile storage device, and another non-volatile storage device such as a hard disk drive may be used. The controller DRAM **120** is a volatile memory using a semiconductor, that is, a dynamic random access memory.

[0032] The controller DRAM **120** stores a CPU setting table **121**, a protocol setting table **122**, an IO processing program **123**, a network setting program **124**, a channel board setting program **125**, and a session management table **126**. However, data and programs stored in the controller DRAM **120** are read from a ROM (not illustrated) at the time of activation. The first controller **130** includes one or more CPUs **131**. The CPU**131** develops a program code stored in a ROM (not illustrated) in the controller DRAM **120** and executes the program code.

[0033] The IO processing program **123** writes data to the SSD **110** and reads data from the SSD **110** based on an operation command received from the host **99** via the smart NIC **200**. The storage controller **100** receives a protocol data unit (PDU) according to a predetermined communication protocol from the smart NIC **200**, and operates based on an operation command included in the PDU. The predetermined communication protocol is a communication protocol that is publicly known or that will be developed in the future, and is iSCSI, Infini Band, NVMe over TCP, or the like. The PDU is also referred to as a network packet, a network frame, communication data, or the like.

[0034] The network setting program **124** generates a session between the host **99** and the storage controller **100** and updates the session management table **126**. The CHB setting program **125** provides a setting screen to an operator, and creates or updates the CPU setting table **121** and the protocol setting table **122** based on an operator's input. Hereinafter, the storage controller **100** and at least one smart NIC **200** are collectively referred to as an information communication device **2**.

[0035] FIG. **2** is a diagram illustrating examples of the CPU setting table **121**, the protocol setting table **122**, and the session management table **126** stored in the controller DRAM **120** of the storage controller **100**.

[0036] The CPU setting table **121** stores settings for a CPU core **132** mounted on each smart NIC **200**. Specifically, the CPU setting table **121** includes a channel board ID, a queue handling dedicated mode, a queue handling dual mode, a CPU resource for port 0, and a CPU resource for port 1.

[0037] The queue handling dedicated mode and the queue handling dual mode are settings for operating the CPU core **132** in one of the queue handling dedicated mode and the queue handling dual mode. The queue handling dedicated mode and the queue handling dual mode have a front and back relationship in which one is set to "Yes" and the other is set to "No". The CPU resource for port 0 and the CPU resource for port 1 are settings for allocating the CPU core **132**. The sum of the CPU resource for port 0 and the CPU resource for port 1 is less than or equal to 100%.

[0038] In the protocol setting table **122**, a communication protocol for each port is set for each smart NIC **200**. Specifically, for each channel board ID, it is described whether each protocol and each port can be applied. In the example illustrated in FIG. **2**, it is described, for a smart NIC **200** of which the channel board ID is "C1", that port 0 supports only "iSCSI Target", and port 1 supports only "NVMe over TCP". Each port is not limited to supporting one protocol, and for example, for a smart NIC **200** of which the channel board ID is "C3", port 1 supports both "iSCSI Target" and "NVMe over TCP".

[0039] Data for each session is stored in the session management table **126**. Specifically, a session ID, protocol setting information, network setting information, and LUN mapping information are included. The session ID is a session identifier, and is expressed by a combination of "E" and a number in the present embodiment. The protocol setting information is setting data related to a communication protocol to be used. Although "iSCSI setting" is described in FIG. **2** for simplification of description, specific setting data is actually stored. The network setting information is setting data related to a network. Although "TCP setting, IP setting" is described in FIG. **2** for simplification of description, specific setting data is actually stored. The LUN mapping information is an identifier of a logical unit number to be used.

[0040] FIG. **3** is a configuration diagram of the smart NIC **200**. The smart NIC **200** includes an NIC ASIC **210**, a DRAM **220**, a second controller **230**, and an embedded memory **240**. The smart NIC **200** includes one or more communication ports, and there is no upper limit on the number of communication ports. However, in the present embodiment, a configuration in a case where the smart NIC **200** has two communication ports, port 0 and port 1, will be specifically described. The smart NIC **200** supports a plurality of network protocols, and can set an IF mode for each communication port to simultaneously support at least the same number of network protocols as the ports. For

example, "iSCSI Target" can be set to port 0 and "NVMe over TCP" can be set to port 1, or "RoCE V2" can be set to both ports.

[0041] The NIC ASIC **210** is an integrated circuit for a specific application that executes processing necessary for operating the NIC, that is, an application specific integrated circuit. The NIC ASIC **210** performs processing independent of a network protocol or processing common to a plurality of network protocols. The DRAM **220** is a volatile memory using a semiconductor, that is, a dynamic random access memory.

[0042] The DRAM **220** stores a smart NIC operation protocol mode table **221**, a parser program **222**, a CPU setting table **223**, a file writing program **224**, a command processing program **226**, a command conversion program **227**, a protocol processing program **228**, a queue handling program **229**, a connection management table **251**, a task management table **252**, and a CPU management table **253**. The protocol processing program **228** is a general term, and actually, a program supporting a communication protocol processed by each port of the smart NIC **200** is stored in the DRAM **220**.

[0043] The protocol processing program **228** is at least one of an iST program **2281**, an NVMe program **2282**, an iSI program **2283**, and an Ro program **2284**. The iST program **2281** is a program that processes "iSCSI Target", which is a communication protocol. The NVMe program **2282** is a program that processes "NVMe over TCP", which is a communication protocol. The iSI program **2283** is a program that processes an "iSCSI initiator", which is a communication protocol. The Ro program **2284** is a program that processes "RoCE V2", which is a communication protocol.

[0044] For example, in a case where "iSCSI Target" is set to port 0 and "NVMe over TCP" is set to port 1, the iST program **2281** and the NVMe program **2282** are activated as the protocol processing program **228**. In addition, in a case where "RoCE V2" is set to both ports, one or two Ro programs **2284** are activated as the protocol processing program **228**.

[0045] The second controller **230** includes one or more CPU cores **231**. Each CPU core **231** is all or some of a plurality of physical CPUs having one or a plurality of arithmetic cores. Although a specific example in which the second controller **230** includes eight CPU cores **231** will be described in the present embodiment, the second controller **230** may include at least one CPU core **231**. The CPU core **231** develops a program code stored in the embedded memory **240** into the DRAM **220** and executes the program code. In the present specification, a program stored in the embedded memory **240** before being executed is referred to as a "program code", and a program code developed in the DRAM **220** is referred to as a "program". For example, two copies of one program code may be created, and two programs for executing the same processing may be arranged in the DRAM **220**.

[0046] Referring to the IF mode file **243** and the smart NIC operation protocol mode table **221**, the parser program **222** activates at least one of a plurality of protocol-based program codes, for example, the iST program **2281**. In a case where the IF mode file **243** does not exist, a specific protocol-based program code is activated based on the description of the smart NIC operation protocol mode table **221**.

[0047] The command processing program **226** is a program that performs processing that can be performed inside the smart NIC **200**. The command processing program **226** performs, for example, processing of initiating or terminating connection. For example, in a case where a command included in the PDU received from the host **99** is reading data stored in the SSD **110**, the smart NIC **200** cannot process the command. Therefore, the PDU is converted into a storage interface command (hereinafter also referred to as a "storage I/F command") and transmitted to the storage controller **100**. However, in a case where the command can be processed by the smart NIC **200**, the command is processed by the command processing program **226**, not transmitted to the storage controller **100**.

[0048] The command conversion program **227** converts the command included in the PDU. Specifically, the command conversion program **227** rewrites the command included in the PDU received from the host **99** into a general-purpose storage interface command that can be processed by the storage controller **100**. In addition, the command conversion program **227** rewrites a command included in the storage interface command received from the storage controller **100** to a command of another communication protocol that can be processed by the host **99** to which a PDU is transmitted. Furthermore, the command conversion program **227** also adds and deletes a task ID and a connection ID to and from the PDU. The queue handling program **229** performs data transfer between a virtual queue and a real queue to be described below. The queue handling program **229** will be described in detail below.

[0049] The embedded memory **240** is a non-volatile storage device embedded in the smart NIC **200**. The embedded memory **240** stores first firmware **240**A, second firmware **240**B, and an IF mode file **243**. However, as will be described below, the IF mode file **243** is not stored in the embedded memory **240** in a shipped state of the smart NIC **200**.

[0050] FIG. **4** is a detailed configuration diagram of the embedded memory **240**. The embedded memory **240** includes a plurality of versions of firmware for a firmware update, and only one version of firmware is used at a time. The first firmware **240**A and the second firmware **240**B have substantially the same configuration, but differ in, for example, versions of program codes to be stored. The IF mode file **243** is a text file as illustrated in the lower part of FIG. **4**, in which a protocol for each port is described. The configuration of the first firmware **240**A will be described below.

[0051] The first firmware **240**A includes an iST program code **2281-0**, an NVMe program code **2282-0**, an iSI program code **2283-0**, an Ro program code **2284-0**, a parser program code **222-0**, a protocol common program code **241**, and setting data **242**.

[0052] Each of the iST program code **2281-0**, the NVMe program code **2282-0**, the iSI program code **2283-0**, and the Ro program code **2284-0** is developed in the DRAM **220** to operate as the iST program **2281**, the NVMe program **2282**, the iSI program **2283**, and the Ro program **2284**. The parser program code **222-0** is developed in the DRAM **220** to operate as the parser program **222**.

[0053] The protocol common program code **241** is developed in the DRAM **220** to operate as the command processing program **226**, the command conversion program **227**, the queue handling program **229**, and the file writing

program **224**. The setting data **242** is developed in the DRAM **220** at the time of activation to become a smart NIC operation protocol mode table **221** and a CPU setting table **223**. Note that the connection management table **251**, the task management table **252**, and the CPU management table **253** are created after activation.

[0054] FIG. **5** is a diagram illustrating examples of the smart NIC operation protocol mode table **221** and the CPU setting table **223** stored in the DRAM **220** of the smart NIC **200** of which the channel board ID is "C1". The smart NIC operation protocol mode table **221** stores correspondence between a content of the IF mode file **243** and an IF mode to be set in the smart NIC **200**. As shown in the first to fourth entries, in principle, the network protocols described after "Protocol=" in the IF mode file **243** are set. In a case where the IF mode file **243** does not exist or in a case where other contents are described in the IF mode file **243**, "iSCSI" is set as shown in the fifth and sixth entries. However, the protocol used in a case where the IF mode file **243** does not exist or in a case where other contents are described in the IF mode file **243** is not limited to "iSCSI", and the operator can set any protocol in the smart NIC operation protocol mode table **221**.

[0055] The CPU setting table **223** stores settings for the second controller **230**. Since the storage controller **100** sends an operation command when the smart NIC **200** is initialized based on a CPU setting list **232**, the values for the corresponding channel board ID in the CPU setting list **232** are identically described in the CPU setting table **223** of the smart NIC **200**. Specifically, the values for the channel board ID "C1" in the CPU setting list **232** shown in FIG. **2** are identically described in the CPU setting table **223** of FIG. **4**.

[0056] FIG. **6** is a diagram illustrating examples of the connection management table **251**, the task management table **252**, and the CPU management table **253** stored in the DRAM **220** of the smart NIC **200** of which the channel board ID is "C1". Connection data for each port is stored in the connection management table **251**. An example of the connection management table **251** for port 0 is illustrated in FIG. **6**. That is, although not illustrated in FIG. **6**, there is also a connection management table **251** in which connections generated using port 1 are described.

[0057] Specifically, a connection ID, a core number, connection information, and a task ID are stored in the connection management table **251**. The connection ID is an identifier for identifying a connection, and is expressed by a combination of "N" and a number in the present embodiment. The task ID is generated each time an IO is received from the host **99**. Since it is common to receive a plurality of PDUs in one connection, the connection ID and the task ID have a one-to-multiple relationship.

[0058] The core number is an identifier of a core constituting the second controller **230**, and is expressed by a combination of "R" and a number in the present embodiment. The connection information is various data related to a connection. Although "Connection setting" is described in FIG. **6** for simplification of description, specific connection data is actually stored. The task ID is a list of identifiers of tasks associated with the corresponding connection, and is expressed by a combination of "T" and a number in the present embodiment. In FIG. **6**, the task ID is described in hexadecimal, but may be described in decimal or the like.

[0059] The task management table **252** stores a state for each task. Specifically, PDU1 and PDU2 are stored for each task ID. The CPU management table **253** stores an identifier of a connection and an identifier of a task to be processed for each core of the second controller **230**. Note that the relationship between the core number and the connection ID and the relationship between the connection ID and the task ID are described in the connection management table **251**. Therefore, the CPU management table **253** can be created based on all the connection management tables **251** included in one smart NIC **200**.

[0060] FIG. **7** is a diagram illustrating an outline of processing executed by the smart NIC **200**. The smart NIC **200** relays bidirectional communication between the host **99** and the storage controller **100**. However, FIG. **7** illustrates a state in which communication from the host **99** to the storage controller **100** is relayed. For communication between the host **99** and the smart NIC **200**, various communication protocols such as RoCEv2, iSCSI, and NVMe over TCP are used. For communication between the smart NIC **200** and the storage controller **100**, PCIe is used. The smart NIC **200** supports various communication protocols with the host **9** by software processing of the second controller **230**.

[0061] Upon receiving communication from the host **99**, the smart NIC **200** first performs protocol analysis, for example, PDU header analysis. Next, the smart NIC **200** allocates a command. Specifically, it is determined whether it is required to send a command to the storage controller **100**. When the smart NIC **200** can process the command by itself, the smart NIC **200** transmits a processing result to the host **9**. When it is required to send a command to the storage controller **100**, the smart NIC **200** continues the processing. Next, the smart NIC **200** converts the PDU into a storage interface command. Specifically, the received PDU header is rewritten so that the storage controller **100** can interpret the PDU header, and the PDU payload is also processed if necessary. Next, the smart NIC **200** stores a storage interface command in a queue for transmission to the storage controller **100**. The queue processing will be described below.

[0062] Next, the smart NIC **200** allocates an interface command. Finally, the smart NIC **200** transfers the storage interface command to the controller DRAM **120** of the storage controller **100** by direct memory access (DMA). Note that, in a case where communication from the storage controller **100** to the host **99** is relayed, the flow is opposite to that in FIG. **7**.

[0063] FIG. **8** is a diagram illustrating an example of allocation of the CPU cores **231** in the second controller **230**. The allocation of the CPU cores **231** in the second controller **230** is determined based on the description of the CPU setting table **223** when the smart NIC **200** is initialized, for example, when the power is turned on. However, when the smart NIC **200** is initialized, the CPU setting of the corresponding smart NIC **200** may be copied from the CPU setting list **232** of the storage controller **100** to the CPU setting table **223**.

[0064] In the example illustrated in the upper part of FIG. **8**, all the eight cores included in the smart NIC **200** are allocated to port 0. In this case, commands received from the host **99-1** connected to port 0 are processed, but commands received from the host **99-2** connected to port 1 are not processed. In the example illustrated in the lower part of FIG. **8**, eight cores included in the smart NIC **200** are equally allocated to port 0 and port 1. In this case, commands received from the host **99-1** and commands received from

the host **99-2** are processed in the same ratio. This configuration corresponds to the CPU setting table **223** illustrated in FIG. **5**.

[0065] FIG. **9** is a schematic diagram illustrating processing for each core of the smart NIC **200**. However, FIG. **9** illustrates an operation in the queue handling dual mode. The storage controller **100** has one command reception queue (C-queue) and one response reply queue (R-queue). The command reception queue is a queue that stores a command from the host **99**. The response reply queue is a queue that stores a reply command to the host **99**. Hereinafter, the command reception queue and the response reply queue are also referred to as "real queues".

[0066] Each CPU included in the second controller **230** has a virtual command reception queue and a virtual response reply queue. The virtual command reception queue and the virtual response reply queue are realized by an area secured in the DRAM **220** of the smart NIC **200**. The virtual command reception queue and the virtual response reply queue are, for example, ring buffers. Hereinafter, the virtual command reception queue and the virtual response reply queue are also referred to as "virtual queues". That is, although there is only one real queue, there are as many virtual queues as the number of cores, and thus, it is necessary to merge the queues or divide the queue. The real queue and the virtual queue are separated to absorb the difference in the current configuration in which the number of queues in the storage controller **100** is different from the number of cores in the CPU core **132** of the smart NIC **200**.

[0067] Each core in the CPU core **132** performs reception processing, transmission processing, and queue handling. The reception processing is processing up to storage in the queue, specifically, the virtual command reception queue in the processing described with reference to FIG. **7** when communication from the host **99** to the storage controller **100** is relayed. The transmission processing is processing subsequent to processing of acquiring a storage interface command from the virtual response reply queue when the communication from the host **99** to the storage controller **100** is relayed. The queue handling is to extract storage interface commands from virtual command reception queues of all the cores, merge the storage interface commands, and transmit the merged storage interface command to the real queue of the storage controller **100** by DMA transfer, and divide a storage interface command acquired from the real queue of the storage controller **100** and store the divided storage interface commands in the respective virtual response reply queues by DMA transfer.

[0068] In the smart NIC **200**, the same connection is handled by the same core. The purpose of this is to conserve resources, avoid cumbersome management, and to avoid performance degradation caused by conflicts and mutual exclusion waits between cores. For example, when the core R3 receives a data request from a certain host **99**, the core R3 transmits data acquired from the storage controller **100** as a response thereto.

[0069] FIG. **10** is a diagram for explaining the queue handling dedicated mode and the queue handling dual mode. However, here, as illustrated in the lower part of FIG. **5**, it is assumed that the CPU resource is allocated to each port by 50%. The upper part of FIG. **10** illustrates the queue handling dedicated mode, and the lower part of FIG. **10** illustrates the queue handling dual mode. The difference between the queue handling dedicated mode and the queue handling dual mode is whether the cores are assigned roles.

[0070] In the queue handling dedicated mode illustrated in the upper part of FIG. **10**, a dedicated core for performing queue handling is set for each port. In the present embodiment, since each smart NIC **200** has two ports, port 0 and port 1, two cores are engaged only in queue handling. Therefore, in this case, the remaining six cores have virtual queues and perform reception processing and transmission processing. Focusing on one port, one core performs only queue handling, and three cores perform reception processing and transmission processing.

[0071] In the queue handling dual mode illustrated in the lower part of FIG. **10**, all the cores perform queue handling, reception processing, and transmission processing. In this case, which core performs the queue handling is undefined. For example, the same core may perform all of queue handling, reception processing, and transmission processing for a command from a certain host **99**. In this case, four cores perform queue handling, reception processing, and transmission processing for each port.

[0072] Hereinafter, an arithmetic core that performs the reception processing and the transmission processing is referred to as an "IO core", and an arithmetic core that performs the queue handling is referred to as a "queue core". In the queue handling dedicated mode, the IO core and the queue core are fixed, but in the queue handling dual mode, the IO core and the queue core are fluid. That is, in the queue handling dual mode, one core operates as an IO core at a certain timing, but operates as a queue core at another timing. The same number of virtual queues as the IO cores are prepared, and in the queue handling dedicated mode, the virtual queues in the lower diagram obtained by subtracting the number of queue cores from the number of arithmetic cores included in the CPU core **132** are prepared. In the queue handling dual mode, the number of IO cores varies depending on the timing, but since any core can be an IO core, virtual queues as many as the arithmetic cores included in the CPU core **132** are prepared.

[0073] There is no clear superiority or inferiority between the queue handling dedicated mode and the queue handling dual mode, each having an advantage and a disadvantage. The advantage of the queue handling dedicated mode is that since the core that performs the queue handling is fixed, an exclusion process between the cores for handling the queues is unnecessary, and there is no overhead of the exclusion process. On the other hand, there is a disadvantage that the number of cores that perform IO processing is reduced. The advantage of the queue handling dual mode is that the number of cores for performing IO processing can be increased. The disadvantage of the queue handling dual mode is that an exclusion process between the cores for handling the queue is required.

[0074] FIG. **11** is a flowchart illustrating processing at the time of hardware initial setting of the smart NIC **200**. The smart NIC **200** executes hardware initial setting according to an instruction from the storage controller **100**. For example, when the operator of the smart NIC **200** newly connects the smart NIC **200** to the storage controller **100**, the storage controller **100** instructs the newly connected smart NIC **200** to execute hardware initial setting.

[0075] The smart NIC **200** first activates a parser program in step S**301**. In the subsequent step S**302**, the parser program **222** determines whether an IF mode file **243** exists.

When an IF mode file **243** does not exist, the processing proceeds to step S**303**, and when an IF mode file **243** exists, the processing proceeds to step S**304**. In step S**303**, referring to the smart NIC operation protocol mode table **221**, the parser program activates a network program for the initial setting mode, that is, "iSCSI Target" corresponding to "no file exists" in the example of FIG. **5**, and proceeds to step S**305**.

[0076] In step S**304**, referring to the smart NIC operation protocol mode table **221** based on the description of the IF mode file **243**, the parser program specifies a network protocol for each port, activates a corresponding network program, and proceeds to step S**305**. In step S**305**, the smart NIC **200** performs initial setting for the channel board. In the subsequent step S**306**, the smart NIC **200** notifies the storage controller **100** of completion of activation of the channel board and the IF mode for each port. For example, when a negative determination is made in step S**302**, a notification such as "CHB activation completed, Port 0: iSCSI, Port 1: iSCSI" is transmitted from the smart NIC **200** to the storage controller **100**.

[0077] The storage controller **100** having received this notification transmits a mode change command to the smart NIC **200** when the IF mode does not match the protocol setting table **122**. For example, the storage controller **100** having received the aforementioned notification from the smart NIC **200** of which the channel board ID "C1" transmits "Mode change command, Port 0: iSCSI, Port 1: NVMe" or the like to the smart NIC **200** because port 1 does not match the protocol setting table **122** illustrated in FIG. **2**. Hereinafter, the IF mode included in the mode change command to be set is also referred to as a "designated mode".

[0078] In the subsequent step S**307**, the smart NIC **200** determines whether a mode change command has been received from the storage controller **100**. When the smart NIC **200** determines that a mode change command has been received, the processing proceeds to step S**308**, and when the smart NIC **200** determines that a mode change command has not been received, the processing proceeds to step S**309**. In step S**308**, the file writing program **224** of the smart NIC **200** writes the designated mode for each port described in the mode change command into the IF mode file **243**, and the processing proceeds to step S**309**. In step S**309**, the smart NIC **200** resets the smart NIC **200**, and ends the processing illustrated in FIG. **11**.

[0079] FIG. **12** is a flowchart illustrating processing at the time of normal activation of the smart NIC **200**. The processing illustrated in FIG. **12** is executed when power supply to the smart NIC **200** is simply started while the smart NIC **200** does not receive a special instruction from the storage controller **100**, or after the reset in step S**309** of FIG. **11**. In FIG. **12**, the same processing as that in FIG. **11** is denoted by the same term. First, in step S**401**, similarly to step S**301**, the smart NIC **200** activates a parser program **222**. In the subsequent step S**402**, the IF mode file **243** written in step S**308** of FIG. **11** is read. In subsequent step S**403**, the smart NIC **200** activates a protocol processing program **228** based on the description of the IF mode file **243**. For example, in a case where the IF mode file **243** is that illustrated in the lower part of FIG. **4**, the iST program **2281** and the NVMe program **2282** are activated as the protocol processing program **228**.

[0080] In the subsequent step S**404**, similarly to step S**305** of FIG. **11**, the smart NIC **200** performs initial setting for the channel board. In the subsequent step S**405**, similarly to step S**306** of FIG. **11**, the smart NIC **200** notifies the smart NIC **200** of the completion of activation of the channel board and the IF mode. In the subsequent step S**406**, the smart NIC **200** determines whether a mode change command has been received from the storage controller **100**. When the smart NIC **200** determines that a mode change command has been received, the processing proceeds to step S**408**, and when the smart NIC **200** determines that a mode change command has not been received, the processing proceeds to step S**407**.

[0081] Note that the processing itself in step S**406** is the same as that in step S**307** of FIG. **11**, but has a different meaning. In step S**307** of FIG. **11**, it is assumed that an IF mode file **243** does not exist because the processing is performed at the time of hardware initial setting, and thus, a change command is received from the storage controller **100** unless initial setting values accidentally match the protocol setting table **122**. That is, in most cases, a change command is received and a positive determination is made in step S**307** of FIG. **11**. On the other hand, in FIG. **12**, since an IF mode designated by the storage controller **100** is written in step S**308** of FIG. **11** executed immediately before, no further change command should be received, and a positive determination is made in step S**406** only in a special case such as a write error.

[0082] In step S**407** executed when a negative determination is made in step S**406**, the smart NIC **200** performs network initial setting and ends the processing illustrated in FIG. **12**. In FIG. **11**, the reset (S**309**) is executed regardless of the determination in step S**307**, but in a case where a negative determination is made in FIG. **12**, processing similar to the reset is not executed.

[0083] In step S**408** executed when a positive determination is made in step S**406**, similarly to step S**308** of FIG. **11**, the file writing program **224** of the smart NIC **200** writes the designated mode for each port described in the mode change command into the IF mode file **243**, and the processing proceeds to step S**409**. In step S**409**, the smart NIC **200** performs a process re-boot and returns to step S**401**. Since a positive determination is made in step S**406** in an exceptional situation as described above, when this process determination is made a predetermined number of times in succession, a blocking process may be performed to disable the smart NIC **200**.

[0084] FIG. **13** is a flowchart illustrating processing of the smart NIC **200** when the IF mode of the smart NIC **200** is switched by the user, in other words, mode change processing. In step S**501**, the smart NIC **200** reads the new setting designated by the user and received via the storage controller **100** and the IF mode described in the IF mode file **243**. In the subsequent step S**502**, the smart NIC **200** determines whether or not the new setting and the IF mode read in step S**501** match. When the smart NIC **200** determines that the new setting and the IF mode match, the processing proceeds to step S**506**, and when the smart NIC **200** determines that the new setting and the IF mode do not match, the processing proceeds to step S**503**.

[0085] In step S**503**, the smart NIC **200** rewrites the IF mode file **243** based on the designation of the user. In the subsequent step S**504**, the smart NIC **200** executes a process re-boot. In step S**505**, the smart NIC **200** executes the

normal activation processing illustrated in FIG. **12** and ends the processing illustrated in FIG. **13**.

[0086] FIGS. **14** and **15** are time charts illustrating processing of the storage system **1** in a case where the IF mode of the smart NIC **200** is switched by the user. This series of steps is too long to be included in one diagram, and thus is divided into two diagrams. That is, in each of FIGS. **14** and **15**, the time elapses from the upper part to the lower part in the diagram, and the beginning of FIG. **15** is a later time than the end of FIG. **14**.

[0087] First, the user performs an operation on the storage controller **100** for switching the IF mode of the smart NIC **200** (S**601**). The IF mode switching operation may be realized using an input device (not illustrated) included in the storage controller **100**, or may be performed by uploading a file storing a new IF mode from a terminal used by the user to the storage controller **100**. When the user performs the IF mode switching operation, the storage controller **100** transmits a storage command including data specifying a new IF mode, that is, a mode change command, to the smart NIC **200** (S**602**).

[0088] The smart NIC **200** that has received the mode change command checks a current IF mode (S**603**), and when the designated IF mode is different from the current IF mode, S**604** to S**615** to be described below are performed. Processing when the designated IF mode and the current IF mode match will be described below. In S**604**, the smart NIC **200** edits the IF mode file **243** and writes the designated IF mode. Then, the smart NIC **200** transmits a response indicating that the mode change command has been received to the storage controller **100** (S**605**). The storage controller **100** that has received this response transmits a storage command including instruction of process re-boot to the smart NIC **200** (S**606**). The smart NIC **200** that has received the storage command transmits a storage command including reception of re-boot to the storage controller **100** and executes the process re-boot (S**608**).

[0089] Thereafter, the storage controller **100** continues polling for checking the activation of the smart NIC **200** (S**609**). When activated, the smart NIC **200** activates the parser program **222** (S**610**), and the parser program **222** activates the protocol processing program **228** based on the description of the IF mode file **243** (S**611**). When the channel board initialization processing is completed (S**612**), the activation of the smart NIC **200** is completed (S**613**). The smart NIC **200** writes the IF mode at the time of activation to the register simultaneously with an activation completion flag (S**614**). Then, the storage controller **100** checks the activation completion flag and the IF mode of the smart NIC **200** (S**615**). The description will be continued with reference to FIG. **15**.

[0090] The initialization of the smart NIC **200** is completed by the above-described processing, and the initial setting for the network port of the smart NIC **200** is performed. Specifically, various initial settings according to the protocol modes are executed between the storage controller **100** and the smart NIC **200** (S**616**). After S**602** described above, when the designated IF mode matches the current IF mode, the smart NIC **200** transmits a normal response indicating that it is not necessary to switch the IF mode to the storage controller **100** (S**617**). This processing corresponds to step S**506** in FIG. **13**.

[0091] FIGS. **16** and **17** are time charts illustrating processing at the time of hardware initial setting of the smart NIC **200**. This series of steps is too long to be included in one diagram, and thus is divided into two diagrams. That is, in each of FIGS. **16** and **17**, the time elapses from the upper part to the lower part in the diagram, and the beginning of FIG. **17** is a later time than the end of FIG. **16**.

[0092] First, the smart NIC **200** is powered on together with a hardware initial setting command. The storage controller **100** continues polling for checking the activation of the smart NIC **200** (S**702**). When activated, the smart NIC **200** activates the parser program **222** (S**703**), and the parser program **222** activates the protocol processing program **228** based on the description of the IF mode file **243** (S**704**). The activation of the parser program corresponds to step S**301** in FIG. **11**, and the activation of the protocol processing program **228** corresponds to steps S**302** to S**304** in FIG. **11**.

[0093] When the channel board initialization processing is completed (S**705**), the activation of the smart NIC **200** is completed (S**706**). The smart NIC **200** writes the IF mode at the time of activation to the register simultaneously with an activation completion flag (S**707**, S**306** in FIG. **11**). Then, the storage controller **100** checks the activation completion flag and the IF mode of the smart NIC **200** (S**708**). The description will be continued with reference to FIG. **17**.

[0094] The initialization of the smart NIC **200** has been completed so far. Next, the storage controller **100** and the smart NIC **200** perform network setting that does not depend on the protocol mode (S**709**). Since the IF mode of the smart NIC **200** may be different from the expectation of the storage controller **100**, the setting that does not depend on the network protocol is performed here. Note that this network setting corresponds to the network port initial setting of the smart NIC **200** illustrated in FIG. **15**. Next, the storage controller **100** performs a network check (S**710**), for example, an internal loop-back test. In order to execute this network check, it is necessary to execute network setting (S**709**) that does not depend on a network protocol as a preliminary step.

[0095] When the IF mode of the smart NIC **200** is different from the expected mode, that is, different from the description in the protocol setting table **122**, the storage controller **100** transmits a storage command including data specifying a new IF mode, that is, a mode change command to the smart NIC **200** (S**711**). The smart NIC **200** that has received the mode change command checks the current IF mode (S**712**), and edits the IF mode file **243** to write the designated IF mode (S**713**). Then, the smart NIC **200** transmits a response to the mode change command to the storage controller **100** (S**714**). Thereafter, hard-reset of the smart NIC **200** is executed, and initialization processing is performed at the time of normal activation.

[0096] According to the first embodiment described above, the following advantageous effects can be obtained.

[0097] (1) The smart NIC **200**, which is an interface module, includes an embedded memory **240** that stores a first firmware **240**A, a second firmware **240**B, and an IF mode file **243**, and a CPU core **231** that executes program codes included in the first firmware **240**A and the second firmware **240**B. Each version of the firmware includes a plurality of protocol-based program codes such as an iST program code **2281-0**, and a parser program code **222-0** that realizes a parser program **222** that activates at least one of the plurality of protocol-based program codes based on the description of the IF mode file **243**. Since it is not necessary to switch a

protocol by firmware exchange while supporting a plurality of network protocols, it is possible to minimize downtime. If the protocol-based program code is stored in a different version of firmware for each protocol, the size of each version of firmware is reduced, but firmware exchange is essential for switching the protocol, increasing downtime. In addition, since it is not necessary to prepare separate firmware for each protocol, not only hardware but also software can be unified.

[0098] (2) The smart NIC **200** includes a file writing program **224** that, when a designated mode indicating a protocol of a program code to be executed at the time of activation is designated, rewrites the IF mode file **243** to activate a protocol-based program code that is the indicated protocol in a case where the IF mode file **243** indicates that a protocol-based program code different from the designated mode is activated (S**502**: NO, S**503** in FIG. **13**).

[0099] (3) When the IF mode file **243** is rewritten, the parser program **222** terminates the activated protocol-based program code and activates a protocol-based program code based on the description of the rewritten IF mode file **243** (S**504** in FIG. **13**).

[0100] (4) The smart NIC **200** includes a plurality of communication ports. The parser program **222** determines the number of processors allocated to each of the communication ports based on the CPU setting table **223** created in advance. Therefore, the operation resources can be allocated according to an expected communication amount for each of the communication ports.

[0101] (5) The smart NIC **200** includes a file writing program **224** that writes the IF mode file **243** into the embedded memory **240** at the time of hardware initial setting.

[0102] (6) When the IF mode file **243** does not exist (S**302**: NO in FIG. **11**), the parser program **222** starts a protocol-based program code determined in advance.

[0103] (7) The storage controller **100** includes a smart NIC **200-1**, which can also be referred to as a first interface module, and a storage controller **100** that receives at least one of a read command and a write command via the first interface module.

[0104] (8) The storage controller **100** includes a controller DRAM **120** that stores a protocol setting table **122** indicating a designated mode which is a communication protocol to be used by the smart NIC **200**, and a CHB setting program **125** that rewrites the mode file to activate a protocol-based program code of the designated mode when the IF mode file **243** of the smart NIC **200** indicates that a protocol-based program code of a protocol different from the designated mode is activated (S**711** in FIG. **17**).

First Modification

[0105] In the first embodiment described above, among the programs stored in the DRAM **220**, only the protocol processing program **228** is prepared for each network protocol. However, the command processing program **226**, the command conversion program **227**, and the queue handling program **229** may also be prepared for each network protocol.

Second Modification

[0106] In the first embodiment described above, the storage controller **100** determines whether the modes match at the time of hardware initialization and at the time of normal activation, and notifies the smart NIC **200** of the result as an IF mode switching command. However, the storage controller **100** may notify the smart NIC **200** of an expected IF mode, and the smart NIC **200** may determine whether the IF mode file **243** needs to be rewritten.

Third Modification

[0107] In the first embodiment described above, the CPU setting table **121** is stored in the controller DRAM **120**, and the queue handling mode and the allocation of CPU resources can be set. However, the queue handling mode may be fixed to either the queue handling dedicated mode or the queue handling dual mode, or the allocation of CPU resources may be unchangeable and fixed to, for example, 50%.

Fourth Modification

[0108] In the first embodiment described above, the same connection is handled by the same core. However, the same connection may be handled by different cores. In addition, it is not essential that the second controller **230** has a plurality of CPU cores **231**, and the second controller **230** may have only one CPU core **231**.

Fifth Modification

[0109] In the first embodiment described above, two versions of firmware are stored in the embedded memory **240**. However, the configuration in which two versions of firmware are stored is merely illustrated with a typical firmware update procedure in mind, and only one version of firmware may be stored in the embedded memory **240**. In addition, three or more versions of firmware may be stored in the embedded memory **240**.

Sixth Modification

[0110] In the first embodiment described above, the smart NIC **200** includes port 1 and the port 2, that is, a plurality of communication ports. However, the smart NIC **200** may include only one communication port.

SECOND EMBODIMENT

[0111] A smart NIC according to a second embodiment will be described with reference to FIG. **18**. In the following description, the same components as those in the first embodiment are denoted by the same reference numerals, and differences will be mainly described. The points that are not specifically described are the same as those in the first embodiment. The present embodiment is different from the first embodiment mainly in that a redundant configuration is realized.

[0112] FIG. **18** is an overall configuration diagram of a storage system **1**A according to the second embodiment. A first smart NIC **200-1** and a second smart NIC **200-2** are connected to the storage controller **100**. Hereinafter, the storage controller **100**, the first smart NIC **200-1**, and the second smart NIC **200-2** are collectively referred to as an information communication device **2**A. A first host **99-1** is connected to the first smart NIC **200-1** and the second smart

NIC **200-2**. A second host **99-2** is connected to the first smart NIC **200-1** and the second smart NIC **200-2**.

[0113] The first host **99-1** and the second host **99-2** use different communication protocols. For example, the first host **99-1** uses iSCSI, and the second host **99-2** uses ROCE V2. Each of the first smart NIC **200-1** and the second smart NIC **200-2** includes two ports, and communication protocols corresponding to the first host **99-1** and the second host **99-2**, respectively, are set. That is, a protocol setting table **122** illustrated in the lower part of FIG. **18** is stored in the storage controller **100** in the present embodiment.

[0114] The information communication device **2**A realizes communication redundancy by using two smart NICs **200**. That is, even if a failure occurs in any one of the first smart NIC **200-1** and the second smart NIC **200-2**, communication between the first host **99-1** and the second host **99-2** and the storage controller **100** can be continued.

[0115] According to the second embodiment described above, the following advantageous effects can be obtained.

[0116] (9) The information communication device **2**A includes a first smart NIC **200-1** and a second smart NIC **200-2**, each of which is the interface module according to claim **1**, and a storage controller **100**. Each of the first smart NIC **200-1** and the second smart NIC **200-2** includes two communication ports, and is capable of supporting different communication protocols for the respective communication ports. In the present embodiment, each of the first smart NIC **200-1** and the second smart NIC **200-2** includes two communication ports and thus supports two or more communication protocols. According to the description of the protocol setting table **122**, a combination of communication protocols supported by the first smart NIC **200-1** matches a combination of communication protocols supported by the second smart NIC **200-2**. Therefore, redundancy of communication can be realized using the two smart NICs **200**.

[0117] The configuration of the functional blocks in each of the above-described embodiments and modifications thereof is merely an example. Some functional components illustrated as separate functional blocks may be integrally configured, or a component illustrated in one functional block diagram may be divided into two or more functions. In addition, some of the function of each functional block may be included in another functional block.

[0118] The above-described embodiments and modifications thereof may be combined together. Although various embodiments and modifications have been described above, the present invention is not limited thereto. Other aspects conceivable within the scope of the technical idea of the present invention also fall within the scope of the present invention.

What is claimed is:

1. An interface module comprising:

a non-volatile storage area that stores one or more versions of firmware and a mode file; and

a processor that executes a program code included in the firmware,

wherein each version of the firmware includes:

a plurality of protocol-based program codes; and

an activation unit that activates at least one of the plurality of protocol-based program codes based on description of the mode file.

2. The interface module according to claim **1**, further comprising:

a file rewriting unit that, when a designated mode indicating a protocol of a program code to be executed at the time of activation is designated, rewrites the mode file to activate the protocol-based program code that is the indicated protocol in a case where the mode file indicates that the protocol-based program code different from the designated mode is activated.

3. The interface module according to claim **2**, wherein

when the mode file is rewritten, the activation unit terminates the protocol-based program code which is activated, and activates the protocol-based program code based on description of the mode file which is rewritten.

4. The interface module according to claim **1**, further comprising

a plurality of communication ports,

wherein the processor includes a plurality of processors, and

the activation unit determines the number of processors allocated to each of the communication ports based on a CPU setting table created in advance.

5. The interface module according to claim **1**, further comprising

a file writing unit that writes the mode file into the non-volatile storage area at the time of hardware initial setting.

6. The interface module according to claim **1**, wherein

when the mode file does not exist, the activation unit activates the protocol-based program code determined in advance.

7. An information communication device comprising:

a first interface module including a first non-volatile storage area that stores one or more versions of first firmware and a first mode file; and

a first processor that executes a first program code included in the first firmware,

wherein each version of the first firmware includes:

a first plurality of protocol-based program codes; and

a first activation unit that activates at least one of the first plurality of protocol-based program codes based on a description of the first mode file; and

a storage controller that receives at least one of a read command and a write command via the first interface module.

8. The information communication device according to claim **7**, further comprising:

a storage device that stores a protocol setting table indicating a designated mode that is a communication protocol to be used by the first interface module; and

a channel board setting unit that rewrites the first mode file to activate the first protocol- based program code of the designated mode when the first mode file of the first interface module indicates that the first protocol-based program code of a protocol different from the designated mode is activated.

9. The information communication device according to claim **7**, further comprising

a second interface module including a second non-volatile storage area that stores one or more versions of second firmware and a second mode file; and

a second processor that executes a second program code included in the second firmware,

wherein each version of the second firmware includes:

a second plurality of protocol-based program codes; and

a second activation unit that activates at least one of the second plurality of protocol- based program codes based on a description of the second mode file,

wherein the storage controller receives at least one of a read command and a write command via the first interface module and the second interface module,

each of the first interface module and the second interface module includes a plurality of communication ports, and is configured to support different communication protocols for each of the communication ports,

each of the first interface module and the second interface module supports two or more communication protocols, and

a combination of communication protocols supported by the first interface module and a combination of communication protocols supported by the second interface module match each other.

10. An activation method executed by an interface module including: a non-volatile storage area that stores one or more versions of firmware and a mode file; and a processor that executes a program code included in the firmware,

each version of the firmware including:

a plurality of protocol-based program codes; and

an activation unit that activates at least one of the plurality of protocol-based program codes based on description of the mode file,

the activation method comprising:

the processor operating the activation unit of any one version of the firmware to activate at least one of the plurality of protocol-based program codes based on the description of the mode file.

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